DOCTORAL PROGRAMME ROADMAP · REVISION B

Organisational Culture as the Conversion Mechanism for Innovation and Artificial Intelligence Readiness in Construction Project Environments

A staged execution plan for the doctoral programme, its five research outputs and the CLEAR-DST diagnostic artefact. Every stage carries its instructions, its deliverables and the criterion on which it is judged complete.

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THESIS	The Influence of Organisational Culture on Employee-Driven Innovation in Construction Project Environments
ARTEFACT	CLEAR-DST: Culture, Leadership, Environment, Architecture, Resources — Diagnostic Support Tool
SUPERSEDES	Claude Code Handover, Revision A, 6 September 2026
ISSUED	6 September 2026

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Reading Instructions

What this document is for, who it binds, and which parts are not advisory

This is the standing execution context for the doctoral programme. It supersedes the Revision A handover in full. Its condensed form, in Appendix E, is placed at the root of each repository as the operating instruction; this document is the reasoning behind that instruction, and it is read in full before acting on any part of it.

What is binding and what is not

Binding. The eight standing guardrails in Section 4.3 and the prohibitions in Section 13. These prevent specific failures that are either irreversible or expensive to reverse, and none of them is suspended when the schedule is under pressure. The gates in Section 10.3 are also binding, in the sense that work does not pass a gate because the date arrived.

Directive. The stage instructions in Part Three. Each is written as the thing to do, and departures from them are recorded with a reason rather than made silently. Where a stage instruction turns out to be wrong, this document is amended and reissued; it is not quietly ignored.

Indicative. The timeline in Section 10.2. It is derived from the dependency structure and anchored on one carried date. It must be reconciled against the actual registration date, the funding end date and the submission deadline before it is treated as a plan of record.

Provisional. Every numeric threshold. Thresholds are named in the text where they matter, but the authoritative values live in `config/thresholds.yml` with their sources attached, and the file is what the code reads. Where this document and that file disagree, the file is wrong and is corrected, or this document is wrong and is amended. They are never left to disagree.

How the parts fit together

1. **Part One** states what changed at Revision B and why, then sets out the aim, objectives, questions and the theoretical architecture. Read it once in full. The partition principle in Section 3.2 and the object substitution in Section 3.3 govern every item that is ever written, so they are worth returning to.
2. **Part Two** sets out the repositories, the guardrails and the stack. Read it before touching either repository, and verify the gitignore before the first commit rather than after.
3. **Part Three** is the working section. Five workstreams, thirty-two stages, each with its purpose, its numbered instructions, its deliverables and the gate that closes it. Work from it stage by stage; the deliverable lists double as checklists.
4. **Part Four** carries the sequencing, the gates, the risks, the ethics obligations and the prohibitions. Read Section 11 when something is going wrong and Section 13 when something looks like a shortcut.
5. **The appendices** are working artefacts rather than supporting material. Appendix A governs item writing, Appendix B governs the review appraisal, Appendix C governs the simulation,

Appendix D governs what gets reported, and Appendix E is the file that goes in the repository.

On the citations in this document

Where this roadmap refers to a published method, threshold or critique, the reference is given in Appendix F and every entry there has been checked against a publisher or index record. Entries that could not be confirmed are marked as unverified and must be confirmed before they are carried into a manuscript. Guardrail G1 applies to this document as it applies to every other file in the programme: a citation that has not been verified is a placeholder, not a reference.

> **TWO THINGS TO DO ON FIRST READING**
>
> Reconcile the timeline in Section 10.2 against the registration and funding record, and open the two long-lead items in Section 10.1, which are the ethics application and the access negotiation. Everything else in the programme can be caught up. Those two cannot.

Part One

1 The Restructure

What changed at Revision B, and the reasoning behind it

Revision A organised the programme around a construct it called organisational conversion capacity, with artificial intelligence supplying the urgency. That ordering put a novel and unvalidated construct at the centre of a thesis whose registered title is about organisational culture, and it left the CLEAR instrument measuring one thing while the theory argued another. Revision B inverts the order. Culture is the object of study. Innovation readiness is what culture produces. Artificial intelligence readiness is a specific and currently binding case of innovation readiness rather than a second research object.

1.1 Why the restructure was necessary

Three defects in Revision A justified the rewrite, and each of them would have surfaced at the viva rather than in the writing.

The construct was redundant. Organisational conversion capacity, as defined in Revision A, is difficult to distinguish from realised absorptive capacity, which already has four decades of theoretical development and validated measurement behind it (Cohen and Levinthal, 1990; Zahra and George, 2002). Introducing a new latent variable that overlaps an established one invites the single most damaging examiner question available: what does your construct explain that the existing construct does not? Answering it would have required a discriminant validity study against absorptive capacity measures, which was nowhere in the plan. Revision B removes the problem by not asserting a new construct. Conversion is treated as a mechanism, not as a latent variable to be measured, and the measurement burden falls where the thesis title already places it, on cultural conditions.

The instrument and the theory disagreed. CLEAR measures Leadership, Environment, Architecture and Resources through perceptual survey items administered to project team members. That is a climate instrument in the strict sense: it captures shared perceptions of practices and conditions, not the deeper assumptions that culture theorists reserve the word culture for (Denison, 1996). Revision A described it as a culture instrument and then theorised about capability. Revision B resolves this by stating the measurement claim precisely in Chapter 3 and by defending the perceptual, referent-shift approach on its own terms rather than by assertion.

Artificial intelligence was a framing device rather than a hypothesis. Revision A used artificial intelligence to motivate the work but never derived a testable claim from it. A reviewer would read that as fashion. Revision B converts the framing into two falsifiable propositions: that artificial intelligence capability has no direct effect on changed project practice, and that a cultural profile predicts artificial-intelligence-enabled practice change better than a technology maturity score does. Both are testable with the data already planned, and both are publishable results whichever way they fall.

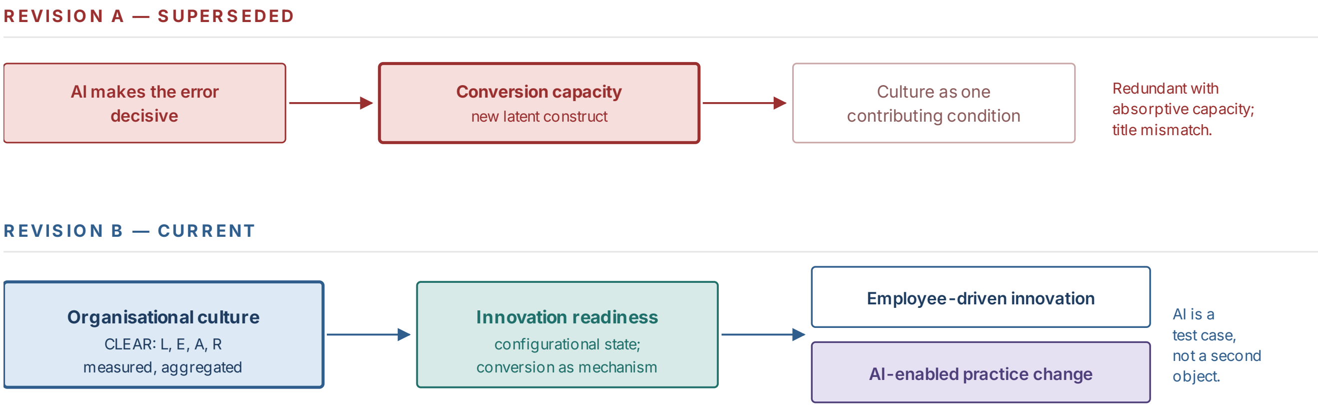


Figure 1 – The programme logic before and after the Revision B restructure. The change is one of ordering, not of subject matter: the same instrument and the same fieldwork now serve a claim the thesis title already commits to.

1.2 The thesis in one paragraph

> **STATEMENT OF THE ARGUMENT**
>
> Construction organisations assess their readiness for innovation and for artificial intelligence by measuring the maturity of the technology they hold rather than the cultural conditions under which people are able to act on it. That measurement error is tolerable while technology changes slowly. Artificial intelligence removes the tolerance, because it raises analytical output sharply while leaving judgement, authorisation and risk acceptance exactly where they were. The binding constraint therefore moves from the availability of analysis to the organisational conditions that decide whether analysis becomes a changed decision. Those conditions are cultural, they are configurational rather than additive, and the condition that binds differs between organisations. A diagnostic that identifies the binding condition is consequently more useful than one that returns a maturity score, and this thesis builds and tests one.

1.3 Schedule of changes from Revision A

Table 1 – Schedule of substantive changes carried at Revision B.

Area	Revision A	Revision B
Central construct	Organisational conversion capacity, a new latent variable to be measured.	Organisational culture, measured through CLEAR. Conversion is a mechanism in the causal argument, not a construct in the measurement model.
Role of AI	Motivating context; no derived hypothesis.	A falsifiable test case. Two hypotheses (H5, H6) turn on it, and one of them pits the cultural profile directly against a technology maturity score.
Measurement claim	Culture instrument, asserted.	Perceptual climate measurement with a referent-shift consensus composition model, defended explicitly in Chapter 3 and tested in Chapter 5.

Area	Revision A	Revision B
Workstreams	Three (review, analysis, artefact). No qualitative workstream, despite RO2 requiring qualitative inquiry.	Five. Construct development (Workstream B) and evaluation and dissemination (Workstream E) are made explicit, closing the gap between the stated mixed-methods design and the build plan.
Dimensional structure	Four conditions of roughly eight items, with a three-condition collapse to be tested.	Four dimensions partitioned on a stated principle (vertical social, horizontal social, structural, material), with the competing three-factor model specified in advance as a rival rather than as a contingency.
Research questions	Three primary questions, none addressing AI.	Four primary questions. RQ3 is new and carries the artificial intelligence argument.
Objectives	Five.	Six. RO3 now covers framework specification separately from instrument construction, because conflating them hid the content validity work.
Outputs	Five papers.	Five papers, unchanged. The prohibition on a sixth output stands.

STANDING PROHIBITION

The restructure does not authorise new scope. The programme still produces five research outputs and one artefact. Anything the restructure appears to open up belongs in the postdoctoral proposal, not in this thesis.

2 Research Architecture

Aim, objectives, questions and the traceability that binds them to the build

2.1 Aim

> **RESEARCH AIM**
>
> To establish how organisational culture governs the conversion of innovative intent and artificial intelligence capability into changed practice in construction project environments, and to instantiate that understanding as a validated diagnostic that identifies the cultural condition binding a given organisation and routes it to the corresponding intervention.

Two clauses do work in that sentence and neither is decoration. Governs the conversion commits the thesis to a mechanism rather than an association, which is what obliges the configurational analysis in Workstream C. Identifies the condition binding a given organisation commits the artefact to a profile rather than a score, which is what forbids an aggregate readiness number anywhere in the tool.

2.2 Objectives

Table 2 – Research objectives at Revision B, with the workstream that discharges each and the output it feeds.

Ref.	Objective	Workstream	Output
RO1	To systematically review and appraise the instruments currently used to measure organisational culture, innovation readiness and artificial intelligence readiness for technology adoption, with particular attention to construction, and to establish what those instruments do and do not measure.	A	Paper 1, Ch. 2
RO2	To establish, through qualitative inquiry with construction practitioners, how cultural conditions enable or constrain employee-driven innovation, and where artificial-intelligence-derived output currently stops in the decision chain.	B	Papers 2 and 3, Ch. 3 and 5
RO3	To specify the CLEAR framework, defining Leadership, Environment, Architecture and Resources as non-overlapping cultural conditions, each mapping to a distinct intervention class, and positioning artificial intelligence readiness as an object-specific case of the same conditions rather than a separate construct.	B	Paper 2, Ch. 3
RO4	To construct and psychometrically validate the CLEAR instrument, establishing structural validity, discriminant validity, measurement invariance across organisation type and firm size, and the defensibility of aggregation from individual responses to a team-level score.	B, C	Paper 3, Ch. 4 and 5

Ref.	Objective	Workstream	Output
RO5	To test the culture-to-innovation relationship both configurationally and structurally, establishing which cultural conditions are necessary, which combinations are sufficient, and whether cultural conversion capacity conditions the effect of artificial intelligence capability on changed practice.	C	Papers 3 and 5, Ch. 5
RO6	To instantiate the validated instrument as the CLEAR-DST artefact, evaluate it in use with practitioners, and derive design principles for cultural diagnosis in project-based organisations that are transferable beyond the software itself.	D, E	Paper 4, Ch. 6

WHERE RO3 CHANGED

Revision A folded framework specification into instrument development. That concealed the content validity work, which is where instrument reviews most often fail. RO3 now stands alone, and Stage B5 exists to discharge it with an expert panel and a quantified content validity index rather than with authorial confidence.

2.3 Research questions

Four primary questions now guide the work. RQ1 and RQ2 are carried forward with sharpened wording. RQ3 is new and carries the artificial intelligence argument. RQ4 is the former RQ3, reframed so that it asks about acting on a diagnosis rather than only producing one.

Table 3 – Primary research questions.

Ref.	Question
RQ1	How does organisational culture shape employee-driven innovation in construction project environments?
RQ2	Which cultural conditions enable or constrain employee-driven innovation, and do they operate as necessary conditions, as sufficient combinations, or both?
RQ3	To what extent does organisational culture condition the conversion of artificial intelligence capability into changed project practice?
RQ4	How can construction organisations diagnose the cultural condition that binds them, and act on that diagnosis?

Table 4 – Research sub-questions, with the analysis that answers each.

Ref.	Sub-question	Answered by
RQ1 · CULTURE AND EMPLOYEE-DRIVEN INNOVATION		
RQ1.1	What is the relationship between psychological safety and employees' willingness to voice innovative ideas in construction projects?	Stage C5 structural model; Stage B4 qualitative mechanism
RQ1.2	How do the structural conditions of project delivery, namely temporariness, contractual relationships and risk aversion, condition the culture-to-innovation relationship?	Stage C5 cross-level moderation; Stage C3 invariance

Ref.	Sub-question	Answered by
RQ1.3	Do vertical cultural conditions, arising from authority, and horizontal conditions, arising among peers, operate through the same mechanism or through different ones?	Stage C5 competing mediation paths
RQ2 · WHICH CONDITIONS, AND HOW THEY COMBINE		
RQ2.1	How does leadership behaviour influence employee-driven innovation, and is its effect direct or transmitted through the peer climate?	Stage C5 mediation; Stage C4 necessity
RQ2.2	How do decision rights and authorisation routes enable or constrain the enactment of employee ideas?	Stage C4 fsQCA; Stage B4 qualitative
RQ2.3	What resource conditions, in time, budget, skill and access to tooling, are necessary to support employee-driven innovation?	Stage C4 necessary condition analysis
RQ2.4	Are there multiple distinct configurations of cultural conditions that produce high innovation readiness, and does the binding condition differ between them?	Stage C4 fsQCA sufficiency and bottleneck analysis
RQ3 · ARTIFICIAL INTELLIGENCE AND THE CONVERSION PROBLEM		
RQ3.1	Where does artificial-intelligence-derived output currently stop in the project decision chain: at analysis, at judgement, at authorisation, or at risk acceptance?	Stage B4 qualitative; Stage C5 outcome decomposition
RQ3.2	Does a cultural profile predict artificial-intelligence-enabled practice change better than a technology maturity score does?	Stage C5 nested model comparison
RQ3.3	Which cultural conditions bind specifically for artificial intelligence output rather than for employee ideas generally?	Stage C4 parallel fsQCA on the AI outcome
RQ4 · DIAGNOSIS AND ACTION		
RQ4.1	Can the cultural conditions be measured to equivalent precision in an adaptive short form rather than the full instrument?	Stage C6 calibration and simulated adaptive study
RQ4.2	Which interventions correspond to which binding condition, and on what evidence?	Stages A5, B4, D5
RQ4.3	Does the diagnosis change what practitioners decide to do, relative to what they would have done without it?	Stage E2 artefact evaluation

2.4 Scope and delimitations

The delimitations carried forward from the thesis draft stand, with two amendments made necessary by the restructure.

Retained. Empirical work is confined to organisations operating in the Irish construction sector, on civil infrastructure and building projects, delivered under traditional, design-and-build and collaborative arrangements. Innovation is treated as employee-led and organisational rather than technical, so specific technologies, tools and materials are not evaluated as innovations in their own right. The analytical focus is internal to the organisation, so regulation and procurement policy enter only where they shape cultural conditions inside projects. Specialist manufacturing, facilities management and organisations outside project-based delivery remain excluded.

Amended, first. Artificial intelligence enters the study as an object that cultural conditions act upon, not as a technology to be evaluated. The thesis makes no claim about the accuracy, capability or appropriate use of any particular system. What it measures is whether artificial-intelligence-derived output reaches a decision, and what determines that. This delimitation must be stated in the methodology chapter, because without it a reviewer will reasonably ask why the study contains no assessment of model performance.

Amended, second. The study measures perceived cultural conditions through a referent-shift consensus model and makes no claim to have measured basic assumptions in the sense used by Schein. Where the thesis uses the word culture it means shared perceptions of the conditions under which discretionary action is possible. That claim is narrower than the title suggests, and stating it plainly in Chapter 3 is cheaper than defending the broader claim in the viva.

DELIMITATION DISCIPLINE

A delimitation is a design decision, not an apology. Each of the above is stated once, in Chapter 1, with the reason attached, and is never re-litigated in the discussion. Limitations, which are different, belong in Chapter 6 with their effect on interpretation named.

3 Theoretical Architecture

How culture, innovation readiness and artificial intelligence readiness relate without construct redundancy

The restructure asks culture and artificial intelligence readiness to sit in one model. The temptation is to add artificial intelligence readiness as a fifth CLEAR dimension. That must be resisted. A fifth dimension would confound an object of adoption with a condition of adoption, would break the partition principle on which the other four rest, and would date the instrument to a single technology. The correct move is narrower and stronger: the four cultural conditions are held constant, and artificial intelligence readiness is what they yield when the object under consideration is machine-derived output rather than a colleague's suggestion.

3.1 The three-layer model

The model has three layers and one crossing arrow. Layer one holds the cultural conditions, measured through CLEAR at individual level and aggregated to the team. Layer two holds the conversion mechanism, in which psychological safety and voice transmit cultural conditions into discretionary behaviour. Layer three holds the enacted outcomes, split into employee-driven innovative work behaviour and artificial-intelligence-enabled practice change. The crossing arrow is the artificial intelligence capability index, which enters at layer three as a moderator rather than at layer one as a condition. That placement is the theoretical claim of the thesis reduced to a diagram.

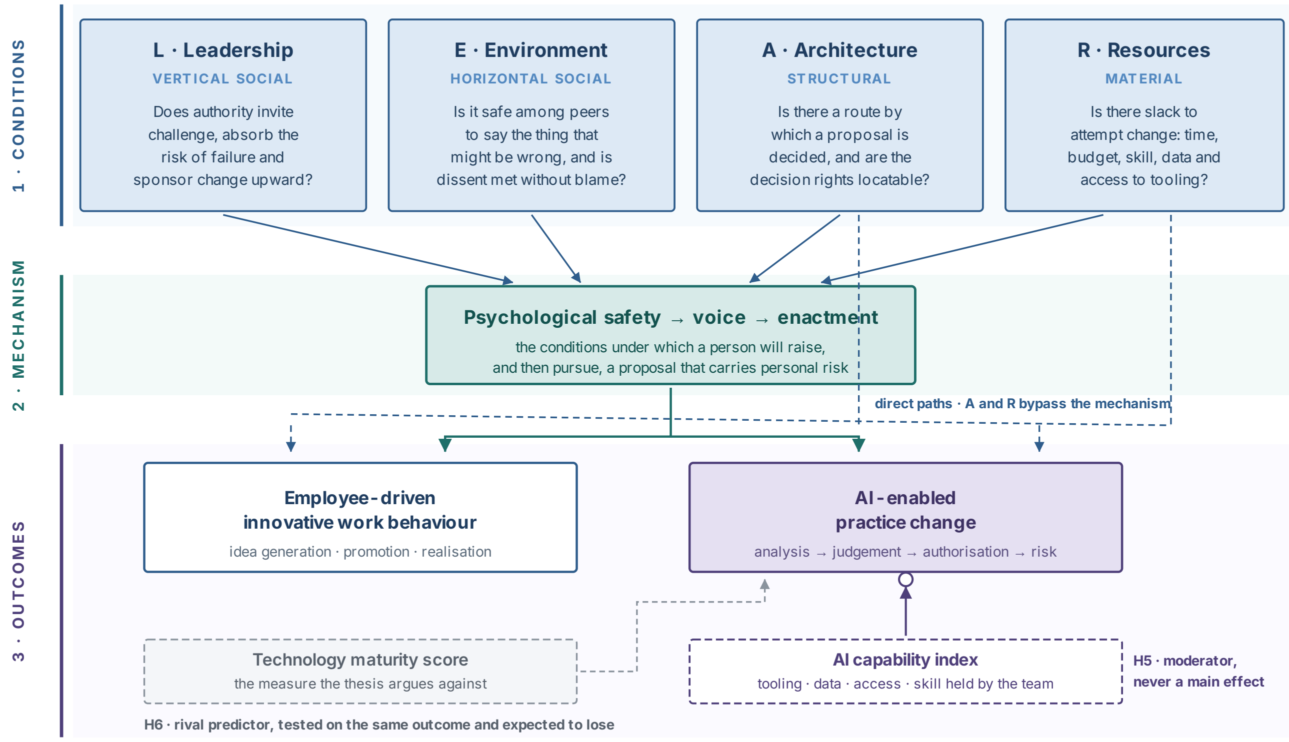


Figure 2 – The Revision B conceptual model. Artificial intelligence capability enters as a moderator of the outcome, not as a cultural condition, and a conventional technology maturity score is retained as a rival predictor so that H6 is a real contest rather than a rhetorical one.

3.2 The partition principle

Four dimensions are defensible only if a reader can say, of any candidate item, which dimension it belongs to and why. CLEAR achieves that through a two-by-two partition that is stated once and then enforced. Two dimensions are social and two are not. Of the social pair, one concerns the relationship with authority and one concerns the relationship among peers. Of the non-social pair, one concerns arrangements and one concerns means. The partition is the reason the instrument has four dimensions rather than three or six, and it is what the confirmatory factor analysis in Stage C2 puts at risk.

Table 5 – The CLEAR partition, with the discriminating question and the intervention class that follows from each condition.

	Dimension	Locus	Discriminating question for item allocation	Intervention class
L	**Leadership**	Social, vertical	Does the item describe the behaviour of someone with authority over the respondent?	Leader behaviour, sponsorship, visible response to ideas, absorption of failure risk
E	**Environment**	Social, horizontal	Does the item describe how peers treat each other, independent of who is in charge?	Team norms, facilitation, blame handling, meeting and review practice
A	**Architecture**	Structural	Does the item describe an arrangement that would persist if every individual were replaced?	Decision rights, authorisation routes, governance and contract interfaces, change control
R	**Resources**	Material	Does the item describe a means whose absence would stop the attempt regardless of willingness?	Time and budget slack, skill and training, data and tooling access, including artificial intelligence tooling

ITEM ALLOCATION RULE

Every item in the pool is allocated by asking the four discriminating questions in the order L, E, A, R and stopping at the first yes. An item that answers yes to two questions is not a borderline item, it is a badly written item, and it is rewritten or discarded at Stage B5. This rule is recorded in the item development log against every item, and the log is an appendix of the thesis.

3.3 Where artificial intelligence readiness sits

Artificial intelligence readiness is not a fifth condition. It is the same four conditions evaluated against a different object. The claim becomes concrete when each dimension is asked about machine-derived output rather than about a colleague's proposal, and the resulting questions are recognisably the ordinary CLEAR questions with the object substituted.

Table 6 – The object substitution. The cultural conditions do not change; what changes is what they are conditions for.

	Condition applied to a colleague's proposal	The same condition applied to machine-derived output
L	Will a manager accept the risk of acting on a suggestion that may prove wrong?	Will a manager accept the risk of acting on a recommendation whose reasoning cannot be fully inspected?
E	Is it safe to contradict a senior colleague's professional judgement?	Is it safe to contradict a senior colleague's judgement *on the basis of* a machine output, and equally safe to reject a machine output that senior colleagues favour?
A	Is there a route by which a proposal becomes an authorised change?	Is there a route by which machine-derived output can enter that authorisation chain at all, and who signs for it?
R	Is there time, budget and skill to attempt the change?	Is there access to the tooling and data, and the skill to interrogate rather than merely receive its output?

Three consequences follow, and each of them is a design decision the programme is now committed to. The instrument does not grow, because artificial intelligence readiness requires no new dimensions. The instrument does acquire a small object-specific block, because the substitutions above are empirical questions and not definitional ones: Stage B5 writes four artificial-intelligence-referenced items, one per dimension, that are administered alongside the main scale and analysed as a separate short form rather than folded into the dimension scores. And the artificial intelligence capability index is measured separately as an inventory of tooling, data access and skill, so that H5 has something to interact with that is not itself a perception.

> **WHY THE OBJECT-SPECIFIC BLOCK IS KEPT OUT OF THE DIMENSION SCORES**
>
> Folding artificial-intelligence-referenced items into the L, E, A and R scores would tie the instrument's calibration to the state of a technology in 2027, and every subsequent administration would drift. Keeping them separate means the core instrument remains valid as the technology moves, and the artificial intelligence block can be re-written without recalibrating the tool. This is the same reasoning that keeps the artefact and the research repositories apart.

3.4 Hypotheses

Six hypotheses are pre-registered. H1 to H4 concern culture and innovation and are the substance of Paper 3. H5 and H6 concern artificial intelligence and are the substance of Paper 5. H6 is the one that decides whether the thesis has a distinctive claim, because it puts the cultural profile into direct competition with the measurement approach the thesis criticises.

Table 7 – Pre-registered hypotheses, with the analysis that tests each and the result that would falsify it.

Ref.	Hypothesis	Test	Falsified if
H1	Team-level cultural conditions predict team-level innovative work behaviour over and above individual dispositional variance.	Multilevel model, Stage C5; variance partitioned before predictors are entered	Level-2 predictors add no explained variance once level-1 controls are in the model

Ref.	Hypothesis	Test	Falsified if
H2	The effect of Leadership on innovative work behaviour is transmitted through the peer climate captured by Environment, whereas Architecture and Resources also act directly.	Multilevel mediation with competing indirect paths, Stage C5; Monte Carlo confidence intervals	The indirect path through Environment is indistinguishable from zero, or the direct paths for A and R vanish
H3	No single cultural condition is sufficient for high innovation readiness; multiple distinct configurations achieve it.	fsQCA sufficiency analysis, Stage C4	A single-condition solution reaches acceptable consistency and coverage on its own
H4	At least one cultural condition is necessary but not sufficient, and the condition that binds differs between organisations.	Necessary condition analysis with bottleneck table, Stage C4; necessity checked before sufficiency	No condition reaches the necessity criterion, or the same condition binds in every case
H5	Artificial intelligence capability has no direct effect on artificial-intelligence-enabled practice change. The effect is conditional on cultural conversion capacity.	Moderation with the capability index, Stage C5; simple slopes across the cultural profile	A direct main effect of capability survives with the interaction in the model
H6	The cultural profile predicts artificial-intelligence-enabled practice change better than a technology maturity score does.	Nested model comparison on the same outcome, Stage C5; incremental fit and out-of-sample criterion	The maturity score matches or exceeds the cultural profile on the pre-specified criterion

THE HONEST POSITION ON H6

H6 can fail, and the pre-registration must say what happens if it does. A maturity score that predicts as well as the cultural profile would not destroy the thesis, but it would demote the artefact from a better instrument to an equally good one that is cheaper to administer. Write that contingency into the pre-registration before the data exist. A hypothesis whose failure has no stated consequence is not a hypothesis.

3.5 Rival explanations to be carried, not avoided

Three rival accounts will be put to the thesis and each is answered by a specific analysis rather than by argument. Naming them now means the analyses exist in the pipeline before anyone asks for them.

1. **Common method variance.** Cultural conditions and innovative behaviour measured from the same respondents at the same moment will correlate for reasons that have nothing to do with the theory. The design answer is a separate informant for the team-level outcome, temporal separation between the predictor and outcome blocks, and a marker variable carried in the instrument from Stage B5 onward. Post hoc statistical correction alone will not persuade a reviewer.
2. **Realised absorptive capacity.** A reader will ask whether the cultural profile is simply absorptive capacity wearing different clothes. The design answer is to carry a short validated absorptive capacity measure alongside CLEAR and demonstrate discriminant validity in Stage C2. This costs eight items and settles the question permanently.

3. **Firm size and project type.** Everything the model attributes to culture might be attributable to the resources of larger firms and the latitude of certain procurement routes. The design answer is measurement invariance testing across firm-size tier and organisation type in Stage C3, and their inclusion as controls in Stage C5.

> **INSTRUMENT CONSEQUENCE**
>
> Two of the three answers above add items to the survey and must be settled at Stage B5, not later. The absorptive capacity short scale and the marker variable go into the instrument at construction. Adding them after the pilot means recalibrating; adding them after fieldwork means not adding them at all.

4 Infrastructure and Standing Conventions

Repositories, guardrails and the technical stack that everything else assumes

No primary data exists yet. Fieldwork opens in the second quarter of 2027, and everything built before then is built against simulated data and swapped over later. That is a design decision rather than a workaround. It removes eighteen months of schedule risk, and it means the analysis code is debugged before the first real response arrives, at which point debugging competes with a submission deadline.

4.1 Two repositories, not one

Research and product are kept apart. Institutional intellectual property in a research repository is a different question from intellectual property in a product repository, and disentangling the two later is considerably harder than separating them now. The only coupling is a single versioned file, and it crosses in one direction.

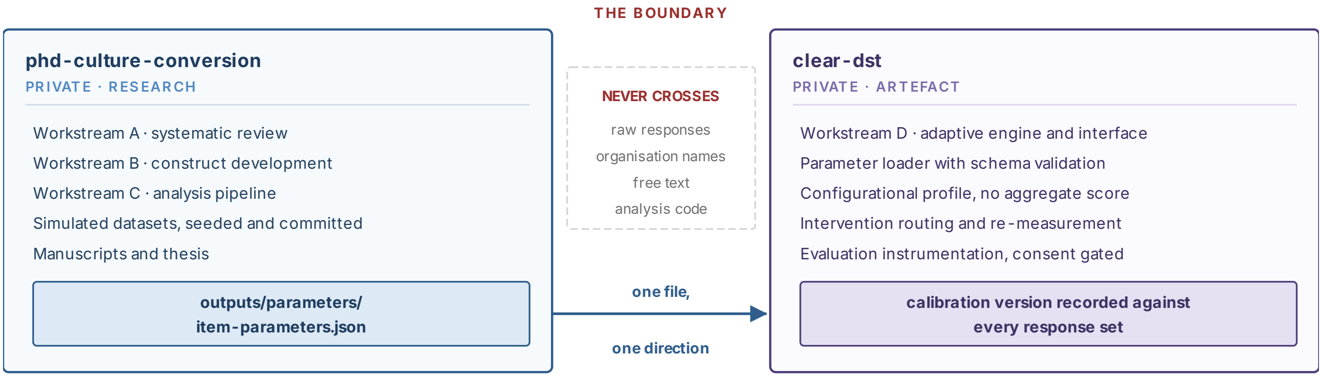


Figure 3 – Repository separation. If a task appears to require both repositories at once, the answer is to export a parameter file, never to merge them.

4.2 Research repository layout

The layout below replaces the Revision A structure. Two directories are new. `construct/` holds Workstream B, which Revision A omitted despite the mixed-methods design requiring it. `config/` holds every analytical threshold in version-controlled files rather than scattered through scripts, so that a decision such as a fuzzy-set calibration anchor is visible in a diff rather than buried in an argument.

```
phd-culture-conversion/                 # private
├── CLAUDE.md                          # standing context, derived from this roadmap
├── renv.lock                          # locked R dependencies
├── _targets.R                         # pipeline orchestration
├── config/
│   ├── thresholds.yml                 # every analytical cutoff, in one place
│   ├── calibration-anchors.yml        # fsQCA anchors, with the reason for each
│   └── seeds.yml                      # one seed per stochastic target
├── data/
│   ├── raw/                           # gitignored at creation, never committed
│   ├── simulated/                     # committed, seeded, reproducible
│   └── derived/                       # gitignored, rebuilt by the pipeline
├── review/                            # Workstream A
│   ├── protocol/                      # registered protocol and any amendments
│   ├── search/                        # strings, database exports, run log
│   ├── screening/                     # decision log, agreement analysis
│   ├── extraction/                    # instrument characteristics
│   └── appraisal/                     # six-criterion rubric scoring
├── construct/                         # Workstream B
│   ├── protocol/                      # interview guide, sampling frame
│   ├── coding/                        # coding frame and analytic memos only
│   ├── item-pool/                     # item development log, allocation decisions
│   ├── content-validity/              # expert panel ratings, CVI computation
│   └── pilot/                         # cognitive interview log, pilot psychometrics
├── analysis/                          # Workstream C
│   ├── simulate/                      # data generator
│   ├── measurement/                   # ordinal CFA, reliability, HTMT
│   ├── invariance/                    # threshold, metric, scalar; aggregation
│   ├── configurational/               # NCA, fsQCA
│   ├── structural/                    # multilevel moderated mediation
│   ├── irt/                           # GRM calibration, CAT simulation, export
│   └── feasibility/                   # design analysis grid for pre-registration
├── outputs/
│   ├── figures/                       # generated only; never hand-edited
│   ├── tables/
│   └── parameters/                    # item-parameters.json crosses to clear-dst
├── manuscripts/                       # Quarto; no hand-typed results
└── thesis/
```

4.3 Standing guardrails

These are absolute. They are not style preferences and they are not negotiable against a deadline. Each is stated with the failure it prevents, because a rule whose purpose is forgotten is a rule that gets suspended.

> **G1 NEVER FABRICATE**
>
> No invented data, results, statistics, citations, digital object identifiers or page numbers, in any file, for any reason, including as a placeholder. Where a number is needed and does not exist, write `TODO_FROM_PIPELINE` and stop. A fabricated placeholder that survives into a manuscript is an academic integrity failure, not a formatting error, and it is not recoverable by explanation after the fact.

G2 NUMBERS REACH MANUSCRIPTS ONLY THROUGH THE PIPELINE

Manuscript files never contain hand-typed results. Values arrive through generated table and figure artefacts or through inline references to pipeline outputs. When a number changes, it changes in one place. This is what makes the manuscripts survive a re-analysis, and a re-analysis is certain: reviewers ask for one.

G3 SIMULATED DATA IS LABELLED AS SUCH, EVERYWHERE, ALWAYS

Every simulated dataset, every figure built from one and every table derived from one carries `SIMULATED` in the filename and in the object metadata. There must exist no path by which a simulated result can be mistaken for an empirical one, including by a co-author, a reviewer, or the candidate eighteen months from now. The label is removed only when the file is deleted.

G4 NO PARTICIPANT DATA IN VERSION CONTROL, EVER

`data/raw/` is gitignored at creation, before any file lands in it, and the ignore is verified before the first commit rather than after. Organisation names, individual names, interview transcripts and free-text responses do not enter either repository in any form. Pseudonymised identifiers only. Workstream B is the highest-risk stage for this rule, because transcripts are text files and text files look harmless.

G5 EVERYTHING IS SEEDED AND REPRODUCIBLE

Every stochastic operation takes an explicit seed drawn from `config/seeds.yml`. The whole pipeline rebuilds from raw inputs with a single command. A result that cannot be regenerated does not exist and is not reported.

G6 EVERY THRESHOLD LIVES IN ONE FILE

Fit index cutoffs, discriminant validity thresholds, agreement criteria, fuzzy-set calibration anchors, consistency and coverage thresholds, adaptive stopping rules: all of them are declared in `config/thresholds.yml` with the source that justifies each, and are read from there by the code. A threshold that appears as a literal inside a script is a threshold nobody will be able to defend under questioning.

G7 THE ARTEFACT NEVER DISPLAYS AN AGGREGATE READINESS SCORE

The absence of a single number is a theoretical commitment, not an oversight. An aggregate score would permit exactly the substitution the thesis argues against, in which a composite figure stands in for knowing which condition binds. This rule survives every usability complaint it attracts, and it will attract some.

G8 BRITISH ENGLISH THROUGHOUT

Code comments, documentation, output labels, item wording and manuscripts. Analyse, behaviour, organisation, modelling, centre, programme. The instrument is administered to Irish respondents and inconsistent orthography in a survey item is a small credibility cost paid on every page.

4.4 Technical stack

Table 8 – Stack, with the reason each component is fixed rather than chosen later.

Layer	Component	Why it is fixed now
Orchestration	`targets`	Declares the dependency graph so that changing one input invalidates exactly the downstream results and nothing else. Without it, reproducibility depends on remembering what to re-run.
Dependencies	`renv`	Locks package versions. Several of the packages below have changed interface across recent major versions, and an examiner will ask which version produced a given result.
Factor models	`lavaan`, `semTools`	Ordinal estimation, invariance testing including the ordinal threshold step, and reliability coefficients appropriate to categorical indicators.
Reliability	`psych`, `MBESS`	Categorical omega with confidence intervals. Coefficient alpha alone will not satisfy a psychometrically literate reviewer.
Item response theory	`mirt`	Graded response calibration, item and test information, and simulated adaptive administration in the same package as the calibration.
Necessity	`NCA`	Ceiling line estimation, effect sizes and bottleneck tables.
Set-theoretic	`QCA`	Calibration, truth table construction, minimisation and the robustness protocols the method now requires.
Reporting	`quarto`	Inline references to pipeline objects, which is what makes guardrail G2 enforceable rather than aspirational.
Artefact	TypeScript, Next.js App Router, PostgreSQL, Prisma	Type safety across the parameter contract, and a relational store for longitudinal re-measurement, which a document store would make awkward.

VERIFY BEFORE ASSUMING

Package names and interfaces are checked against current releases at the point of setup rather than assumed from memory. Several of the packages above have changed function signatures across major versions, and the failure mode is silent: an argument that has been renamed is ignored, the model fits, and the result is wrong in a way nothing warns about.

5 Workstream A: Evidence Base

A PRISMA 2020 systematic review of how culture, innovation readiness and artificial intelligence readiness are currently measured

6	A1	P1	B5
STAGES	STARTS IMMEDIATELY	FEEDS PAPER 1, CHAPTER 2	BLOCKS ITEM WRITING

This workstream does two jobs at once, and the second is the one that matters more. Its visible product is a published systematic review. Its structural product is the item pool inheritance and the justification for building a new instrument rather than adopting an existing one. If the review finds an adequate instrument already validated for this purpose, the programme changes shape and it is far cheaper to discover that in 2026 than in 2028.

Scope: instruments that claim to measure organisational culture, climate for innovation, innovation readiness, digital readiness or artificial intelligence readiness in organisational settings, published from 2005 to the search date, with construction and engineering applications extracted separately. Appraisal is against a six-criterion measurement rubric derived from established measurement-property taxonomies rather than from a rubric invented for the occasion.

THE QUESTION THE REVIEW MUST ACTUALLY ANSWER

Not what instruments exist. That produces a catalogue and catalogues are not publishable. The question is: *what do these instruments measure the maturity of, and what do they leave unmeasured*? The expected finding, which the review must be capable of contradicting, is that the literature measures the presence of technology and the existence of policy, and does not measure the organisational conditions under which either becomes a changed decision. Design the extraction form so that this claim can be evidenced item by item, not asserted in the discussion.

A1 Protocol, rubric and registration

Blocks all of A · No dependencies

The protocol is written and registered before the first search runs. Registration is not a formality: it is what separates a systematic review from a narrative review with a flow diagram, and reviewers at the target journals increasingly check.

INSTRUCTIONS

1. Write the review question in a structured form, naming the population (organisations and teams in project-based and comparable settings), the construct (culture, climate for innovation, innovation readiness, artificial intelligence readiness), the instrument type (self-report questionnaires and structured maturity assessments), and the measurement properties to be appraised.
2. Specify inclusion and exclusion criteria in operational terms, so that two screeners applying them independently reach the same decision without discussion. Vague criteria are the single

largest source of screening disagreement, and disagreement is expensive at full-text stage.

3. Define the six appraisal criteria and their scoring anchors in full before any record is read. Appendix B carries the rubric; it is reproduced verbatim in the protocol. The criteria are construct definition, theoretical anchoring, object of measurement, level of analysis, psychometric evidence, and predictive evidence.
4. Specify the dual-screening plan: what proportion of records is screened by two people, at which stages, and which chance-corrected agreement statistic is reported. Fix the acceptable threshold in the protocol rather than after seeing the result.
5. Register the protocol. Check the eligibility rules of the health-focused registers first, and where the review falls outside their scope, register on an open repository with a timestamped, non-editable record. Record the registration identifier in `review/protocol/`.
6. Pre-specify the synthesis approach: narrative synthesis structured by measurement property, with the appraisal matrix as the primary result. State that meta-analysis is not appropriate and why, so that a reviewer does not ask for one.

KEY DELIVERABLES

- [ ] Registered protocol with a public identifier and date, stored in `review/protocol/`
- [ ] Six-criterion appraisal rubric with anchors and worked examples for each criterion
- [ ] Operational inclusion and exclusion criteria, screener-testable
- [ ] Dual-screening plan with a pre-committed agreement threshold
- [ ] Second screener identified and briefed, with the briefing recorded

GATE A1

Two people independently screen the same twenty pilot records against the draft criteria and reach acceptable agreement without conferring. If they do not, the criteria are the problem, not the screeners. Revise and repeat before proceeding.

A2 Search design and execution

Depends on A1

A search is a reproducible artefact, not an activity. Every string, every database, every filter and every date is recorded such that a third party could re-run the search and obtain the same yield within the drift introduced by database updates.

INSTRUCTIONS

1. Build the search string from three concept blocks joined with AND: the construct block, the measurement block, and the setting block. Within each block, terms are joined with OR and include the variants the literature actually uses, which are more numerous than expected. Innovation climate, innovation readiness, absorptive capacity, digital readiness, artificial intelligence readiness, technology readiness, maturity model and organisational capability all appear for the same underlying idea.
2. Develop the string against a known-item set: assemble ten to fifteen papers that must be retrieved, and iterate until the string returns all of them. A string that misses a known item is a string that is missing a term, and known-item testing is the only way to find out which one.

3. Search a minimum of two large multidisciplinary databases with different coverage profiles, plus at least one discipline-specific source for construction and engineering. Coverage overlap between the large indexes is substantial but incomplete, and searching only one is a criticism that costs nothing to avoid.
4. Record for each database: the exact string, the fields searched, the filters applied, the date of the search, and the number of records returned. Export in a structured format and preserve the export file untouched alongside the parsed output.
5. Supplement the database search with backward and forward citation chasing on the included set, and record what each supplementary method contributed. Hand-searching that is not recorded cannot be reported.
6. Write the parser as code. Manual reformatting of exports is not reproducible and introduces errors that are invisible afterwards.

KEY DELIVERABLES

- [ ] Search strings for every database, in full, ready for reproduction in an appendix
- [ ] Untouched raw exports in `review/search/` , alongside the parsed record table
- [ ] Search run log: database, string, fields, filters, date, yield
- [ ] Known-item test result showing full recall of the seed set
- [ ] Supplementary search log with the contribution of each method

GATE A2

The parser runs from the raw exports to the record table with no manual step, and the record count it produces matches the sum of the yields in the run log.

A3 Deduplication and record management

Depends on A2

Deduplication is where PRISMA counts most often become indefensible, because silent merges cannot be audited afterwards.

INSTRUCTIONS

1. Match on digital object identifier first, as an exact match. Records with identical identifiers are merged automatically and the merge is logged.
2. Apply fuzzy matching on normalised title plus year to the remainder. Every near-match is routed to a review queue for a human decision. Nothing is merged silently on a similarity score.
3. Log every merge decision with both record identifiers, the matching method, the similarity value where applicable, and the decision. The log is a file in the repository, not a state inside a reference manager.
4. Retain the pre-deduplication record table. The PRISMA figure needs both counts and reconstructing the earlier one later is not possible.

KEY DELIVERABLES

- [ ] Deduplicated record table, with a stable identifier per record that persists through every later stage
- [ ] Merge decision log, complete and machine-readable
- [ ] Pre-deduplication and post-deduplication counts, derived from the data

A4 Screening and agreement

Depends on A3

One row per record per stage. That structure is what makes the flow diagram derivable and the process auditable, and it costs nothing to adopt at the start and a great deal to retrofit.

INSTRUCTIONS

1. Build the screening log with one row per record per stage, carrying: record identifier, stage, decision, reason code, screener, and timestamp. Reason codes come from a closed list defined in A1, because free-text exclusion reasons cannot be counted.
2. Screen titles and abstracts against the inclusion criteria, erring toward retention. A record wrongly excluded at this stage is never seen again; a record wrongly retained costs one full-text read.
3. Dual-screen the pre-specified subsample independently, resolve disagreements by discussion, and record how each disagreement resolved. Compute and report the chance-corrected agreement statistic named in the protocol.
4. Retrieve full texts for everything surviving. Records that cannot be retrieved are reported as such with a count, not quietly dropped.
5. Screen full texts, recording a single primary exclusion reason per excluded record. The reason distribution is itself a reportable result, and a large count against one reason usually indicates a criterion that was ambiguous.
6. Generate the flow diagram from the screening log. It is derived, never drawn. A hand-drawn flow diagram will eventually disagree with the record table and the disagreement will be found by a reviewer.

KEY DELIVERABLES

- [ ] Complete screening log reconstructing every decision by record, stage, screener and reason
- [ ] Agreement analysis on the dual-screened subsample, with the statistic and its interpretation
- [ ] Generated flow diagram, reproducible from the log by a single command
- [ ] Full-text exclusion table with reasons and counts

GATE A4

Regenerating the flow diagram from the screening log reproduces every count in the manuscript with no manual step, and agreement on the dual-screened subsample meets the threshold pre-committed in A1. Agreement below threshold is reported rather than concealed, with the criteria revision that followed.

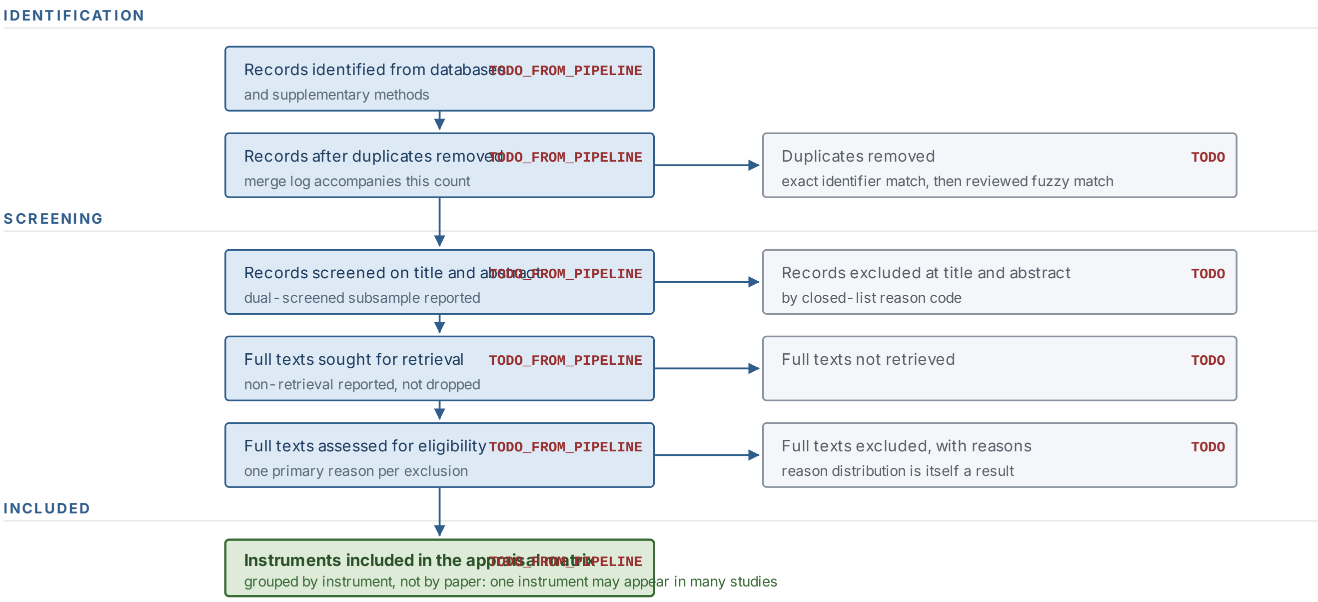


Figure 4 – Flow diagram template for Workstream A. Every count is a placeholder until the screening log supplies it, and the diagram is generated from that log rather than drawn. A count typed by hand here is a guardrail G1 violation.

A5 Extraction and measurement appraisal

Depends on A4 · Blocks B5

Extraction is where the review earns its contribution. A form that captures only bibliographic characteristics produces a catalogue. A form built around the six criteria produces an argument.

INSTRUCTIONS

1. Extract at the level of the instrument, not the paper. One instrument may appear across many studies with different evidence attached, and the appraisal is of the instrument as it now stands with all its accumulated evidence.
2. For each instrument capture: the construct label used by its authors, the construct definition given, the theoretical source, the object of measurement (technology, policy, capability, perception or behaviour), the intended level of analysis, the level at which it was validated, the number of items and dimensions, the response format, the sample and setting, and the reported measurement evidence.
3. Score each instrument against the six criteria using the anchors from A1. Every score carries an evidence field quoting or citing exactly what justified it. A score without evidence is an opinion and will be treated as one.
4. Extract the object of measurement with particular care, because this field carries the review's central claim. Record whether items ask about the presence of a technology, the existence of a policy, a perceived condition, or an observed behaviour, and count them.
5. Record the level-of-analysis mismatch explicitly: instruments claiming to measure an organisational property from individual responses without an aggregation justification are a specific and countable defect, not a general weakness.
6. Dual-extract a subsample and report agreement on the appraisal scores, not only on the screening decisions. Appraisal is more subjective than screening and is where a reviewer will probe.

7. Extract the item stems of any instrument whose licence permits it, into `review/extraction/`. This becomes the inherited item pool for Stage B5 and is the single most valuable by-product of the review.

KEY DELIVERABLES

- [ ] Extraction table, one row per instrument, with every field populated or explicitly marked as not reported
- [ ] Scored appraisal matrix across the six criteria, with an evidence field on every cell
- [ ] Object-of-measurement tabulation, item-counted, supporting or refuting the review's central claim
- [ ] Level-of-analysis defect count with the instruments named
- [ ] Inherited item pool with provenance and licence status recorded per item
- [ ] Dual-extraction agreement on appraisal scores

INTEGRITY CONDITION

The review must be capable of concluding that an adequate instrument already exists. If the appraisal is designed such that no existing instrument can pass, it is not an appraisal, it is a justification. Write the anchors so that a hypothetical instrument could score well on all six criteria, and say in the protocol what the programme would do if one did.

A6 Synthesis and reporting

Depends on A5 · Delivers Paper 1

The synthesis is organised by measurement property rather than by instrument or by chronology. Organising by instrument produces a list; organising by property produces the comparative claim that makes the review useful.

INSTRUCTIONS

1. Build the corpus characteristics table from the record table by code, covering publication years, settings, sectors, instrument types and sample sizes.
2. Structure the narrative synthesis by criterion. For each of the six, state what the corpus does well, what it does badly, and what the pattern implies for anyone who intends to use these instruments to make a decision.
3. Write the construction sub-analysis as a distinct section. Whether construction-specific instruments differ from general organisational instruments is a question practitioners in the field will ask first.
4. Report the artificial intelligence readiness sub-corpus separately, including whether those instruments have any psychometric validation at all as distinct from face-valid maturity grids.
5. Write the gap statement as a specific, evidenced claim about what is unmeasured, tied to the tabulated object-of-measurement counts. Position the programme's instrument against that gap in the final paragraph and nowhere earlier.
6. Produce the reporting artefacts through the pipeline: flow diagram, appraisal matrix, corpus characteristics table, and the completed reporting checklist required by the target journal.

KEY DELIVERABLES

- [ ] Paper 1 manuscript, with every count and table generated rather than typed
- [ ] Completed PRISMA 2020 checklist, including the abstract checklist
- [ ] Appraisal matrix as a publishable figure or table
- [ ] Corpus characteristics table
- [ ] Gap statement, evidenced from the extraction data
- [ ] Chapter 2 draft derived from the same source files

GATE A6 · WORKSTREAM A DEFINITION OF DONE

Running the pipeline from the raw database exports reproduces every count, figure and table in the manuscript with no manual step, and the screening log reconstructs every decision. Until that is true, the review is a draft regardless of how finished the prose looks.

6 Workstream B: Construct Development

From practitioner accounts to a pilot-tested instrument, by way of an auditable item development log

7	RO2	A5	C
STAGES	DISCHARGES THE QUALITATIVE OBJECTIVE	INHERITS THE ITEM POOL	BLOCKS THE ENTIRE ANALYSIS PIPELINE

Revision A had no qualitative workstream, which left objective RO2 undischarged and the item pool arriving from nowhere. That is the most common way an instrument-development thesis fails: the items appear in Chapter 4 with no traceable origin, and the examiner asks where they came from. Workstream B makes the origin auditable. Every item in the final instrument can be traced backward through a content validity rating, an allocation decision, a source, and in most cases a practitioner utterance.

> **TWO THINGS THIS WORKSTREAM IS NOT**
>
> It is not a study in its own right competing with Papers 3 and 4 for space, and it is not grounded theory. It is instrument development conducted with qualitative methods, reported as such. The interviews serve the construct and the item pool. Where they produce a finding worth publishing independently, that finding goes into Paper 2 as theoretical development, not into a sixth output.

B1 Access, sampling frame and ethics

Blocks B3 · Long lead time, start early

Access is the longest-lead item in the whole programme and the one least under the candidate's control. It is started first and treated as a schedule risk rather than an administrative task.

INSTRUCTIONS

1. Build the sampling frame across the three organisational forms the thesis names: contractors, consultants and client organisations. Within each, stratify by firm size tier, because firm size is a rival explanation the design must be able to answer, and by role level, because vertical and horizontal cultural conditions look different from different heights in an organisation.
2. Target purposive maximum variation rather than convenience. A sample drawn entirely from the candidate's existing network will be criticised, and correctly. Record how each participating organisation was approached and by whom.
3. Draft the participant information sheet and consent form for both the interview phase and, in the same submission, the later survey phase and the artefact evaluation. One ethics application covering three phases is far faster than three sequential applications, and amendments are cheaper than new submissions.
4. Address the employer-employee hazard explicitly in the ethics submission. Access is granted by management while data is given by employees, which creates a coercion risk that a generic consent form does not answer. State that participation is not reported to the employer, that

individual responses are never returned to the organisation, and that team-level reporting is suppressed below a stated minimum number of respondents.

5. Write the data management plan: what is collected, where it is stored, who has access, how long it is retained, and the point at which identifiers are destroyed. Distinguish pseudonymised from anonymised data honestly, and do not claim anonymisation for data that retains a re-identification key.

6. Secure a signed access agreement with each participating organisation covering both phases, so that survey recruitment does not require re-negotiation at the point when the schedule has no slack.

KEY DELIVERABLES

- [ ] Sampling frame with stratification by organisational form, size tier and role level
- [ ] Ethics approval covering interviews, survey and artefact evaluation in one application
- [ ] Participant information sheets and consent forms for each phase
- [ ] Data management plan with retention and destruction schedule
- [ ] Signed access agreements, covering both empirical phases
- [ ] Access log recording every approach and its outcome, including refusals

GATE B1

Ethics approval granted and access agreements signed for enough organisations to make the Stage C7 feasibility target reachable. Interviewing before approval is not a shortcut, it is unusable data.

B2 Interview protocol and pilot

Depends on B1

The protocol is built to elicit conditions and incidents, not opinions about culture. Asking practitioners to describe their organisational culture produces the vocabulary of management training. Asking them to describe the last time they raised an idea and what happened produces data.

INSTRUCTIONS

1. Structure the guide around critical incidents. Ask for a specific occasion on which the participant, or someone they observed, proposed a change to how work was done. Then trace the incident forward: who heard it, what they did, where it stopped, and what the participant concluded from the outcome.

2. Cover all four CLEAR conditions without naming them. The interview should be capable of surfacing a resource constraint, an authorisation dead end, a leader's reaction and a peer's reaction without the participant being told those are the four categories.

3. Include an artificial intelligence block that mirrors the same incident structure: the last time a machine-derived output was produced or received, what was done with it, and where it stopped. The four-step chain of analysis, judgement, authorisation and risk acceptance is the analytic frame, and RQ3.1 is answered from these accounts.

4. Include a disconfirming probe in every section. Ask for an occasion when the opposite happened. A guide that only elicits confirmation produces a coherent and unpersuasive

analysis.

5. Pilot the guide with two or three participants who are not in the main sample. Revise timing, sequencing and wording, and record the revisions with reasons.
6. Fix the recording, transcription and storage chain before the first interview: where recordings live, how they reach transcription, when they are destroyed, and how pseudonyms are assigned and stored separately from the transcripts.

KEY DELIVERABLES

- [ ] Interview guide, versioned, with the pilot revisions and their reasons recorded
- [ ] Artificial intelligence incident block, mapping to the four-step decision chain
- [ ] Pseudonymisation protocol with the key held separately from the data
- [ ] Recording, transfer, transcription and destruction chain, documented and tested

B3 Fieldwork

Depends on B2 · Highest data-protection risk in the programme

INSTRUCTIONS

1. Interview across the stratified frame, monitoring coverage as fieldwork proceeds rather than after it. A frame that ends up all main contractors and no clients is a problem that can only be fixed while interviewing is still running.
2. Write an analytic memo within twenty-four hours of each interview, recording what was surprising, what contradicted the emerging account, and what should be probed differently next time. Memos are data and are committed to the repository; transcripts are not.
3. Judge sufficiency against the purpose. The purpose is item generation and mechanism identification, so the stopping criterion is that new interviews stop producing new conditions or new stopping points in the decision chain, not a target count fixed in advance. Record the point at which that was judged and on what basis.
4. Transcribe to a consistent standard. Verbatim with speaker turns is sufficient; conversation-analytic notation is not needed for this purpose and costs a great deal of time.
5. Strip identifiers at transcription, not later. Organisation names, project names, individual names and any detail that identifies a project by its characteristics are replaced at the point the transcript is made.

GUARDRAIL G4 APPLIES MOST SHARPLY HERE

Transcripts never enter version control. The repository holds the coding frame, the analytic memos and the aggregate coding output. Verify the ignore rule before the first transcript exists, not after. A transcript committed and later removed remains in the history, and rewriting shared history is not a remedy anyone should have to attempt under submission pressure.

KEY DELIVERABLES

- ☐ Completed interviews across the stratified frame, with a coverage table
- ☐ Pseudonymised transcripts, stored outside version control per the data management plan
- ☐ Analytic memo set, committed
- ☐ Sufficiency judgement recorded with its basis

B4 Analysis and mechanism identification

Depends on B3 · Feeds Paper 2 and RQ3.1

Coding is abductive: the CLEAR conditions supply an initial template, and the data is permitted to break it. A template analysis that never revises its template has not analysed anything.

INSTRUCTIONS

1. Build the initial template from the four conditions plus the four-step decision chain, then code a subset and revise the template against what the subset actually contains. Record every template revision with the case that prompted it.
2. Code specifically for stopping points. For each incident, identify where the proposal or the machine output stopped, and what stopped it. The distribution of stopping points across the corpus is a primary result and directly answers RQ3.1.
3. Test the partition principle empirically. Attempt to allocate every coded condition to exactly one of L, E, A and R. Conditions that resist allocation are reported, and if there are many, the four-dimension structure is in trouble and it is better to know now than after calibration.
4. Have a second coder independently code a subsample and report agreement. Where a second coder is not available, conduct and document a structured negative case analysis instead, and say which was done.
5. Write the mechanism account: how a cultural condition becomes, or fails to become, an enacted change. This account is the substance of Paper 2 and of Chapter 3, and it is what turns the framework from a taxonomy into a theory.
6. Extract candidate item content. Every distinct condition or stopping point that appeared is a candidate for an item, and the utterance that produced it is recorded as its provenance.

KEY DELIVERABLES

- ☐ Final coding template with a complete revision history
- ☐ Stopping-point distribution across the four-step decision chain, tabulated
- ☐ Partition test result, including conditions that resisted single allocation
- ☐ Coder agreement or documented negative case analysis
- ☐ Mechanism account, drafted as the core of Chapter 3
- ☐ Candidate item content with utterance-level provenance

B5 Item generation and content validity

Depends on A5, B4 · The stage instrument reviews attack

Two sources feed the item pool: the inherited items from the review, and the candidate content from the interviews. Both are logged with provenance. What leaves this stage is not a set of items, it is a set of items each of which has a documented reason to exist and a quantified expert judgement attached.

INSTRUCTIONS

1. Assemble the pool at roughly three times the intended final length per dimension. Writing thirty candidate items to keep ten is normal and is far cheaper than discovering at pilot that a dimension has too few items that work.
2. Apply the item allocation rule from the partition principle: ask the four discriminating questions in order and stop at the first yes. Record the allocation and the question that settled it against every item.
3. Write every item with a consistent referent. The referent is the team, not the individual and not the organisation, because the composition model is referent-shift consensus and mixing referents inside a scale breaks the aggregation argument before it is made.
4. Write the four artificial intelligence object-specific items, one per dimension, using the object substitutions in Part One. These are administered with the instrument and scored separately.
5. Add the carried measures the rival explanations require: a short validated absorptive capacity scale, a marker variable unrelated to the constructs, and the established psychological safety and innovative work behaviour scales. Use published scales for these rather than writing new ones, because the point of carrying them is comparability.
6. Run a formal content validity exercise. Recruit a panel combining measurement expertise and construction domain expertise, have each member rate every item for relevance to its assigned dimension and for clarity, and compute item-level and scale-level content validity indices. Retain, revise or discard on the pre-committed threshold rather than on the candidate's judgement.
7. Conduct a separate discriminant sorting exercise: give a second panel the four dimension definitions and the unlabelled items, and ask them to sort. An item that a majority allocate to the wrong dimension is a defective item regardless of its relevance rating.
8. Record everything in the item development log: source, provenance, allocation, allocating question, content validity index, sorting result, decision and reason. This log is an appendix of the thesis and the single most effective defence against the question of where the items came from.

KEY DELIVERABLES

- [ ] Item pool at roughly three times target length, fully provenanced
- [ ] Item development log, complete, one row per item with every decision recorded
- [ ] Content validity indices at item and scale level, against a pre-committed threshold
- [ ] Discriminant sorting results with misallocation rates per item
- [ ] Artificial intelligence object-specific block, four items, one per dimension
- [ ] Carried measures selected and licensed: absorptive capacity, psychological safety, innovative work behaviour, marker variable
- [ ] Instrument version 0.1 assembled for cognitive interviewing

NOTHING IS ADDED AFTER THIS STAGE

Every measure the analysis needs must be in the instrument when it leaves B5. The absorptive capacity scale and the marker variable exist solely to answer rival explanations, and if they are omitted here they cannot be recovered later at any price. Check the analysis plan for Stages C2 to C5 against the instrument, item by item, before the instrument is frozen.

B6 Cognitive interviewing and instrument refinement

Depends on B5

Cognitive interviewing establishes that respondents understand each item as intended. It is cheap, it takes a fortnight, and skipping it is the reason many instruments produce uninterpretable factor structures that are then explained away as sample characteristics.

INSTRUCTIONS

1. Recruit eight to twelve respondents spanning the roles and organisational forms the instrument will meet in the field, including at least one site-based role and one senior role.
2. Use concurrent think-aloud with targeted verbal probes. Ask what the item is asking, what the respondent thought of when answering, and how they chose between adjacent response options.
3. Probe the referent explicitly on a sample of items. If respondents answer team-referenced items about themselves or about the whole company, the composition model fails and the wording must change.
4. Probe the artificial intelligence block hardest. Terminology in this area is unstable and a respondent's understanding of the phrase may differ from the candidate's. Establish what respondents include and exclude, and word the items to match practice rather than to match the literature.
5. Record every comprehension failure against the item, revise, and re-test the revised items with fresh respondents rather than the same ones.
6. Check completion time. An instrument that takes longer than fifteen minutes will suffer both non-response and satisficing in a site environment, and both damage the data in ways that cannot be corrected afterwards.

KEY DELIVERABLES

- [ ] Cognitive interview log, one entry per item per respondent
- [ ] Comprehension failure register with the revision made against each
- [ ] Referent verification result
- [ ] Measured completion time distribution
- [ ] Instrument version 0.2, ready for pilot administration

B7 Pilot administration and item reduction

Depends on B6 · Supplies pilot parameters to Workstream D

The pilot is not a small version of the main study. Its purpose is to reduce the item pool to the instrument that will be fielded, and to supply the provisional item parameters that let the artefact be built before the main data exists.

INSTRUCTIONS

1. Administer the full pool to a pilot sample large enough to examine item behaviour, drawn from organisations that will not be in the main sample. Reusing organisations contaminates the main study and complicates the invariance argument.
2. Examine item-level distributions first. Items with severe ceiling effects, minimal variance, or an unused response category are candidates for removal before any model is fitted. The Revision A note that the pilot showed response compression makes this step mandatory rather than routine.
3. Examine inter-item and item-total relationships within each intended dimension, and cross-loadings between dimensions. An item correlating as strongly with a neighbouring dimension as with its own has failed the partition and is removed.
4. Fit the exploratory structure. This is the pilot's job and it belongs here, not in the main study, where the analysis is confirmatory. Report the pilot as exploratory and the main study as confirmatory, and never blur the two.
5. Reduce to the target instrument length, retaining items that span the content of the dimension rather than the items with the highest loadings. A dimension represented by ten near-identical items has high reliability and narrow content, and it will fail the content validity argument that B5 worked to establish.
6. Fit provisional graded response parameters on the pilot data and export them as `item-parameters.json` with a calibration version marked clearly as provisional. Workstream D consumes this file and builds against it.
7. Re-run the completion time check on the reduced instrument and confirm it meets the target.

KEY DELIVERABLES

- [ ] Pilot dataset, pseudonymised, held outside version control
- [ ] Item performance table: distribution, variance, category use, correlations, cross-loadings
- [ ] Exploratory structure result with the retention decisions it drove
- [ ] Reduced instrument version 1.0, frozen for fieldwork
- [ ] Content coverage check confirming each dimension spans its definition
- [ ] Provisional `item-parameters.json` , versioned and labelled provisional
- [ ] Item reduction log with the reason for every removal

GATE B7 · WORKSTREAM B DEFINITION OF DONE

A frozen instrument whose every item traces to a source, an allocation decision, a content validity rating and a pilot performance record; a provisional parameter file exported and consumed successfully by the artefact repository; and no route by which an item could have entered the instrument without leaving a trace in the log.

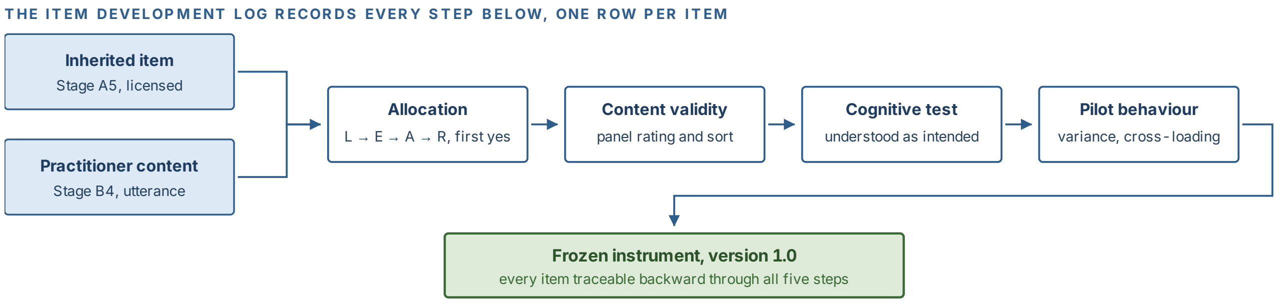


Figure 5 – Item provenance chain. An item that cannot be traced backward along this chain does not enter the instrument, whatever its psychometric performance.

7 Workstream C: Measurement and Analysis Pipeline

Built, debugged and proven against simulated data before a single real response exists

7
STAGES

C1
BLOCKS EVERYTHING ELSE

2
WORLDS TO RECOVER: FOUR-FACTOR AND THREE-FACTOR

C7
FEEDS THE PRE-REGISTRATION

The whole of this workstream is written and tested before fieldwork opens. That is the single most valuable decision in the programme. Code written against real data under submission pressure is code written once and checked never, and every analytical decision made after seeing the data is a decision a reviewer is entitled to treat as opportunistic. Writing it first converts the analysis plan from a promise into an executable artefact.

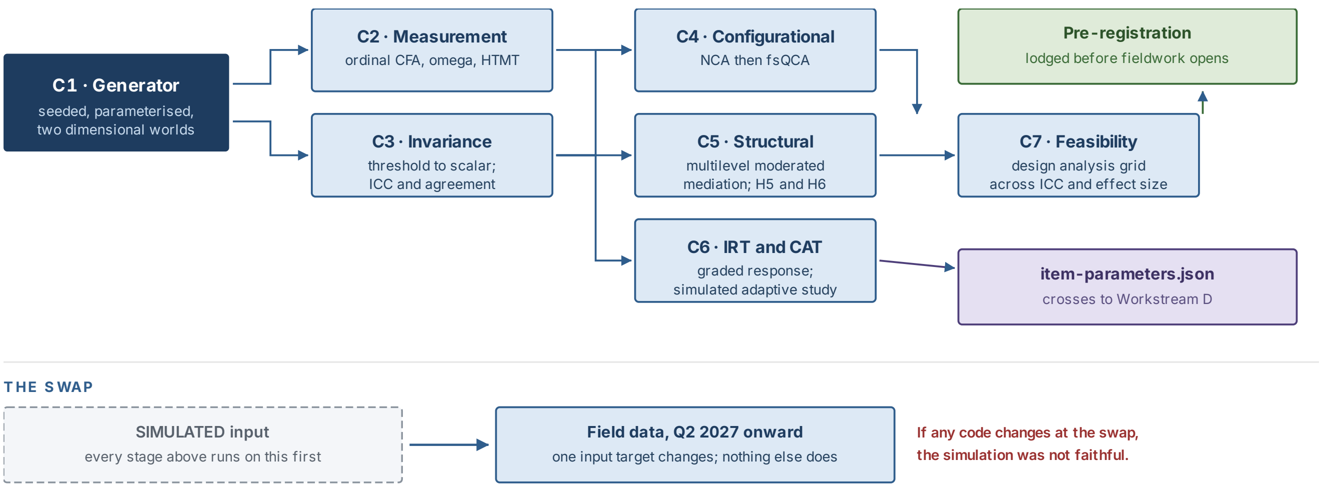


Figure 6 – The analysis pipeline as a dependency graph. The swap from simulated to field data changes one input target and nothing else; any code change required at that moment is evidence that the generator did not match the planned design.

C1 The data generator

Build this first · Blocks C2 to C7

The generator produces a nested dataset matching the planned design exactly: teams of four to five members, four dimensions of ten items, five-point ordinal responses, plus the artificial intelligence capability index, the carried absorptive capacity and psychological safety scales, the innovative work behaviour measure, and a team-level outcome from a separate informant.

INSTRUCTIONS

1. Parameterise the intraclass correlation so that aggregation can be tested at plausible values rather than at a single optimistic one. Sweep the realistic organisational range and observe what happens to the aggregation statistics at each value.

2. Parameterise the dimensional structure with two worlds: a four-condition world matching the CLEAR partition, and a three-condition world in which two dimensions collapse. Both must be generated and both must be correctly recovered by the Stage C2 code. Building only for the

four-factor case leaves the programme with no tested code on the day the four-factor model fails.

3. Parameterise the effect sizes on the culture-to-innovation relationship, on the mediation paths, and on the artificial intelligence interaction, so that the feasibility grid in C7 has something to vary.
4. Model response compression and social desirability explicitly, since the pilot showed both. Compression is generated by shrinking the usable response range; social desirability by an additive respondent-level shift correlated with a marker. Code that has never seen compressed responses will behave unexpectedly when it meets them.
5. Model missingness and partial team completion. Real teams will not all return four members, and the aggregation and multilevel code must be exercised against teams of two, three and five before it meets them for real.
6. Attach metadata to every generated object recording the generating parameters, the seed and the word `SIMULATED`, and carry that metadata into every downstream figure and table automatically rather than by convention.
7. Write the generator's own tests: for a large sample at known parameters, the recovered parameters must match the generating parameters within a stated tolerance. A generator that is not itself verified silently invalidates every stage that depends on it.

KEY DELIVERABLES

- [ ] Generator function with a documented parameter interface, seeded throughout
- [ ] Four-condition and three-condition generating worlds, both exercised
- [ ] Compression, social desirability, missingness and partial completion switches
- [ ] Parameter recovery test at large sample, passing within tolerance
- [ ] Committed simulated datasets in `data/simulated/`, each labelled and reproducible from its seed

THE THREE-CONDITION WORLD IS NOT A CONTINGENCY

It is a specified rival model, tested against the four-condition model on the same data at Stage C2. Treating it as a fallback to be reached for if the preferred model fails is what produces the appearance of post hoc model fitting. Specify both in the pre-registration, state which is preferred and why, and report the comparison whichever way it falls.

C2 Measurement model

Depends on C1 · Answers RO4

INSTRUCTIONS

1. Treat five-point items as ordinal and estimate accordingly, using a robust weighted least squares estimator with polychoric correlations. Treating five-category items as continuous biases loadings downward and produces fit statistics that are not comparable to the ordinal literature.
2. Fit the four-factor model and the three-factor rival on the same data and compare them. Report both, with the comparison statistic and the decision rule stated in advance in `config/thresholds.yml`.

3. Report robust fit indices, and state in the methods chapter that conventional cutoffs were derived under maximum likelihood with continuous indicators and are contested for categorical estimation. Report the indices, give the cutoffs used, and cite the source of the cutoffs rather than presenting them as neutral facts.
4. Resist modification indices. Every correlated residual added to reach acceptable fit is a departure from the specified model and must be justified substantively and reported as a departure. Fitting by modification index and reporting the result as confirmatory is one of the most common and most easily detected failures in this literature.
5. Report reliability with an omega coefficient appropriate to categorical indicators, with a confidence interval. Coefficient alpha may be reported alongside for comparability with older studies but is not the primary evidence, because its assumptions are not met here.
6. Assess discriminant validity with the heterotrait-monotrait ratio against the threshold pre-committed in the protocol, and report the inferential form as well as the point estimate. The partition principle from Part One is what this test puts at risk, so a failure here is theoretically informative rather than merely inconvenient.
7. Run the discriminant test against the carried absorptive capacity scale as well as between CLEAR dimensions. This is the answer to the construct redundancy objection and it is only available if B5 carried the scale.
8. Assess common method variance using the marker variable carried from B5, not with a single-factor test. Report the marker-adjusted estimates alongside the unadjusted ones.

KEY DELIVERABLES

- ☐ Fitted four-factor and three-factor models with a reported comparison
- ☐ Standardised loadings, residuals and any departures from specification, each justified
- ☐ Categorical omega per dimension with confidence intervals
- ☐ Heterotrait-monotrait matrix with inferential bounds, between dimensions and against absorptive capacity
- ☐ Marker-variable method-variance assessment
- ☐ Recovery demonstration: the code correctly identifies the four-factor world as four-factor and the three-factor world as three-factor

GATE C2

On simulated data from the four-condition world the code selects the four-factor model; on simulated data from the three-condition world it selects the three-factor model. If it cannot distinguish them, the measurement model has no diagnostic value and the instrument needs more discriminating items before fieldwork, not after.

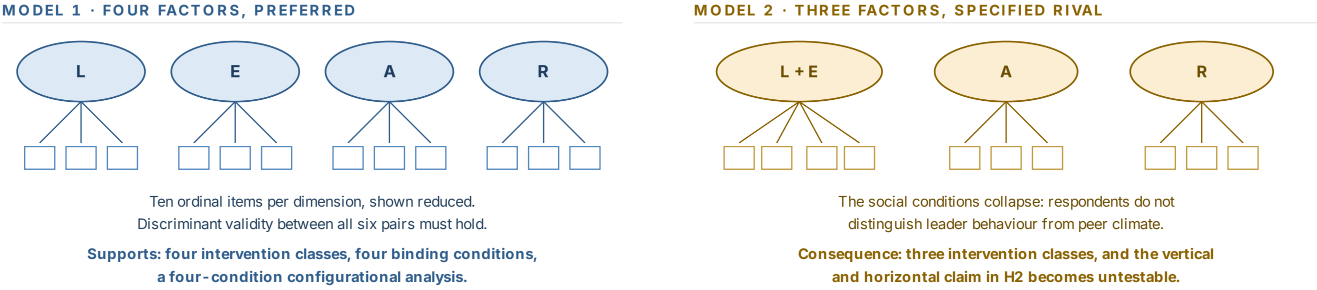


Figure 7 – The two measurement structures the pipeline must be able to tell apart. The three-factor rival is not a lesser outcome, it is a substantive finding with direct consequences for the artefact's intervention routing, and it is specified in advance for that reason.

C3 Invariance and aggregation

Depends on C2 · The design's weakest joint

Aggregation from individual perceptions to a team score is a claim about the data, not a convenience. It has to be earned, and the design makes earning it harder than it looks.

INSTRUCTIONS

1. Test measurement invariance across organisation type and firm-size tier following the sequence appropriate to ordinal indicators, which begins with thresholds rather than with loadings. Using the continuous-indicator sequence on ordinal data tests the wrong constraints in the wrong order.
2. Evaluate each step against the change criteria pre-committed in `config/thresholds.yml`, citing their source. Where full invariance fails, establish partial invariance, name the non-invariant parameters, and state exactly what comparisons remain permissible.
3. Compute ICC(1), ICC(2) and within-group agreement for every dimension, and report all three. ICC(1) speaks to whether team membership matters; ICC(2) to whether the team mean is reliable; agreement to whether members actually agree. They answer different questions and reporting only one is a common and consequential shortcut.
4. Compute within-group agreement under more than one null distribution rather than the uniform null alone, and report the range. A single null assumption produces an agreement figure that is easy to challenge.
5. Report the significance test on between-team variance. Aggregating when teams do not differ is aggregating noise.
6. Run the aggregation diagnostics across the intraclass correlation sweep from C1 and record where they fail, before fieldwork. This tells the programme what the field data must look like for aggregation to be defensible, and that is information worth having in advance.

A STRUCTURAL PROBLEM THE DESIGN MUST CONFRONT NOW

ICC(2), the reliability of a team mean, rises with both the intraclass correlation and the number of respondents per team. With four to five members per team, ICC(2) remains modest across the entire plausible range of intraclass correlation values for organisational perception data. The consequence is that the team means may not reach conventional reliability even when the underlying construct behaves exactly as theorised.

Three responses are available, and the choice is made now rather than after the data arrives. First, increase respondents per team where access permits, which raises ICC(2) directly and is the only response that solves rather than manages the problem. Second, run the configurational analysis at team level as planned but report the structural model with individual-level predictors and cluster-robust standard errors, which does not require reliable team means. Third, report ICC(2) honestly against the interpretive standards and argue the case on ICC(1) and agreement, which is defensible but must be argued rather than assumed. Stage C7 quantifies the trade-off; the decision belongs in the pre-registration.

KEY DELIVERABLES

- [ ] Invariance sequence results across organisation type and firm-size tier, with the criteria and their source
- [ ] Partial invariance findings, with the permissible comparisons stated
- [ ] ICC(1), ICC(2) and agreement per dimension, with the between-team variance test
- [ ] Agreement computed under multiple null distributions, reported as a range
- [ ] Aggregation defensibility map across the intraclass correlation sweep
- [ ] Aggregation decision recorded in the pre-registration, with its fallback

C4 Configurational analysis

Depends on C3 · Tests H3 and H4 · Primary result

The configurational analysis is the primary test, and the structural model complements it rather than replacing it. That ordering follows from the theory: if the claim is that the binding condition differs between organisations, an analysis that estimates one average effect per condition cannot test it.

INSTRUCTIONS

1. Run necessity before sufficiency. A condition that is necessary will appear in every sufficient configuration and will be misread as a driver if necessity is not established separately.
2. Conduct necessary condition analysis with both ceiling line estimators, report effect sizes against the published interpretation bands, and test necessity statistically rather than reading it from a scatter plot. Produce the bottleneck table, which is the output the artefact's intervention routing depends on.
3. Set calibration anchors in `config/calibration-anchors.yml` with a written justification for each of the three anchors on every condition. Anchors set at sample quantiles without a substantive reason are the most frequently criticised element of published set-theoretic work, and the criticism is usually correct.
4. Avoid setting any case exactly at the crossover point, since such cases are dropped from the analysis. Check for and report them.

5. Construct the truth table with a frequency threshold appropriate to the sample size and report how many cases and what proportion of the total the retained configurations cover. Report the extent of limited diversity honestly: with four conditions there are sixteen logically possible configurations, and a sample of this size will not populate all of them.
6. Apply the consistency and proportional reduction in inconsistency thresholds pre-committed in the configuration file, and report all three solution types. Present the intermediate solution as primary with the directional expectations stated in advance, and report the parsimonious solution alongside so that core and peripheral conditions can be distinguished.
7. Run the robustness protocol: vary the calibration anchors, the consistency threshold and the frequency threshold across a stated range, and report how the solution changes. A solution that survives only at one parameter setting is not a finding.
8. Run the parallel analysis on the artificial-intelligence-enabled outcome to answer RQ3.3, and compare which conditions bind for machine-derived output against which bind for employee ideas. Any difference between the two solutions is a genuinely novel result.
9. Analyse the negation of the outcome as well. Configurations sufficient for low readiness are not the mirror image of those sufficient for high readiness, and reporting only one half invites the criticism that causal asymmetry was assumed rather than examined.

KEY DELIVERABLES

- [ ] Calibration file with a written justification for every anchor on every condition
- [ ] Necessity results with effect sizes, statistical tests and the bottleneck table
- [ ] Truth table with frequency and consistency thresholds, and a limited-diversity statement
- [ ] Complex, parsimonious and intermediate solutions, with core and peripheral conditions marked
- [ ] Robustness report across anchors and thresholds
- [ ] Parallel solution on the artificial intelligence outcome, compared against the innovation outcome
- [ ] Solution for the negated outcome
- [ ] Configuration-to-intervention mapping, which Stage D5 consumes directly

C5 Structural model

Depends on C3 · Tests H1, H2, H5, H6

INSTRUCTIONS

1. Begin by partitioning variance. Report the unconditional model and the proportion of variance lying between teams for every outcome, before any predictor enters. This is what makes H1 a real test rather than a formality.
2. Specify the mediation within a multilevel structural equation framework rather than by entering group means as predictors in a single-level model, because the latter conflates within-team and between-team effects and produces indirect effects that are not interpretable at either level.
3. Obtain confidence intervals for indirect effects by Monte Carlo or a comparable resampling method, not from a normal-theory standard error, since the sampling distribution of a product is not symmetric.

4. Test the vertical and horizontal claim in H2 by specifying competing indirect paths in the same model rather than in separate models, so that the comparison is within a single estimated structure.
5. Test H5 as a moderation of the artificial intelligence outcome by the capability index, reporting the interaction and the simple slopes across the range of the cultural profile. Report the main effect of capability with the interaction in the model, because H5 predicts it will not survive.
6. Test H6 as a nested model comparison on the same outcome with the same sample: cultural profile against technology maturity score, then both together. Pre-specify the comparison criterion in the pre-registration, and pre-specify what would count as the maturity score winning.
7. Build the single-level fallback now, not later: design-effect-adjusted analysis with cluster-robust standard errors, ready to run if recruitment falls short of what C7 shows the multilevel model needs. Building the fallback in advance means the decision to use it is a sample-size decision rather than a decision made under duress.
8. Carry firm size and organisation type as controls throughout, and report the model with and without them.

KEY DELIVERABLES

- [ ] Variance partition for every outcome, reported before predictors
- [ ] Multilevel mediation with competing indirect paths and Monte Carlo intervals
- [ ] Moderation results for H5 with simple slopes and the capability main effect
- [ ] Nested comparison for H6 with the pre-specified criterion
- [ ] Single-level fallback, built and tested on simulated data
- [ ] Control-variable sensitivity, models reported with and without

ON REPORTING ORDER

Chapter 5 reports the configurational results first and the structural results second. That order follows the theory, and reversing it would signal that the configurational analysis is a robustness check on a regression, which is the opposite of the argument. Reviewers at construction management journals are more familiar with the structural form, so the chapter must earn the ordering with a short paragraph explaining why the question requires it.

C6 Item response calibration and adaptive simulation

Depends on C2 · Supplies Workstream D · Feasibility risk

This stage carries the largest gap between what the programme has promised and what the sample can support, and confronting that gap now is considerably cheaper than confronting it in the viva.

THE ARITHMETIC THAT HAS TO BE FACED

The artefact is specified to reach equivalent measurement precision in eight to twelve items rather than the full instrument. With four dimensions, each requiring its own ability estimate, eight to twelve items means two or three items per dimension. Two or three graded-response items cannot produce a standard error comparable to that of a ten-item scale, whatever the item selection algorithm does, because the information simply is not there. The claim as written is not achievable and must be restated before it appears in a paper.

Three routes are available. The first is to restate the target: an adaptive form of sixteen to twenty items, four to five per dimension, is still a reduction of more than half and is honestly defensible. The second is to fit a multidimensional graded response model that borrows information across correlated dimensions, which genuinely improves efficiency when the dimensions correlate as CLEAR's are expected to, and which would itself be a methodological contribution in this literature. The third is to accept a lower precision target for screening use and report the standard error attained rather than claiming equivalence. Route two is preferred, route one is the fallback, and route three is what the artefact does in the interim. Whichever is chosen, the target is stated as a standard error, never as an item count.

INSTRUCTIONS

1. Establish the assumptions before fitting: sufficient unidimensionality within each dimension, local independence, and monotonicity. A graded response model fitted to a dimension that is not unidimensional produces parameters that mean nothing.
2. Fit the graded response model per dimension with full item fit diagnostics, and inspect category thresholds for disordering. Disordered thresholds usually indicate a response category that respondents do not use, which is a finding about the instrument and a candidate for collapse.
3. Produce item and test information functions per dimension and identify where on the trait continuum the instrument is precise. An instrument informative only at the extremes is of little use for diagnosis, since organisations near the middle are the ones that need to know which condition binds.
4. Establish the calibration sample requirement against the simulation literature and compare it honestly with what the planned fieldwork will yield. Record the gap in the risk register and state the mitigation.
5. Fit the multidimensional model alongside the per-dimension models and compare the precision achieved at fixed test length. This comparison is the evidence for whichever route is taken.
6. Run the simulated adaptive study: maximum information selection, ability estimation, exposure control, content balancing so that every dimension is represented, and stopping rules expressed as standard error targets with a maximum length. Compare against full-form administration on precision, exposure and the standard error trajectory.
7. Export item parameters as versioned JSON against a published schema, with the calibration version, the sample it was calibrated on, and whether it is provisional or final recorded inside the file rather than in a filename.

KEY DELIVERABLES

- [ ] Assumption checks per dimension, reported rather than assumed
- [ ] Calibrated parameters with item fit statistics and threshold ordering
- [ ] Item and test information functions, with the precision range stated
- [ ] Calibration sample requirement against planned yield, with the gap named
- [ ] Multidimensional versus per-dimension precision comparison at fixed length
- [ ] Simulated adaptive study: precision, exposure, standard error trajectory, length distribution
- [ ] Restated precision target, expressed as a standard error and not an item count
- [ ] `item-parameters.json` against a published schema, versioned

C7 Feasibility and design analysis

Depends on C1 to C5 · Delivers the pre-registration

This stage converts the simulator from scaffolding into a methodological contribution. It establishes what the planned sample can and cannot detect, and it does so before recruitment rather than as a post hoc power calculation, which is not informative.

INSTRUCTIONS

1. Define the grid: number of teams, respondents per team, intraclass correlation, and effect size on each key path. Run the full pipeline across the grid with many replications per cell and record detection rates, parameter bias and interval coverage.
2. Report power for each hypothesis separately. The cross-level interaction in H5 will require substantially more than the main effects in H1, and a single overall power figure conceals that.
3. Report the aggregation surface alongside power: at which combinations of intraclass correlation and team size does the aggregation argument hold, and at which does it fail. This is the output that decides the recruitment target.
4. Report the configurational feasibility separately, since set-theoretic analysis is governed by case numbers and diversity rather than by statistical power. State the minimum number of cases and the expected extent of limited diversity.
5. Derive the recruitment target from the grid rather than from convention, and state the minimum below which the multilevel model is abandoned in favour of the single-level fallback. Put that number in the pre-registration so that the decision is made in advance.
6. Write the pre-registration: hypotheses, measures, sampling, exclusion rules, the full analysis plan for each stage, every threshold, the aggregation decision and its fallback, the H6 comparison criterion, and what happens if each hypothesis fails. Lodge it before fieldwork opens.

KEY DELIVERABLES

- ☐ Design analysis grid results: detection rate, bias and coverage per cell per hypothesis
- ☐ Aggregation defensibility surface across intraclass correlation and team size
- ☐ Configurational feasibility statement with minimum case numbers
- ☐ Recruitment target with the abandonment threshold for the multilevel model
- ☐ Lodged pre-registration with a public timestamp
- ☐ Methods paper material: the simulation study is publishable in its own right within Paper 3 rather than as a sixth output

GATE C7 · WORKSTREAM C DEFINITION OF DONE

A single command runs the full pipeline end to end on simulated data and produces every figure and table the manuscripts need; the code correctly recovers both the four-condition and the three-condition structures from their respective worlds; the feasibility grid has produced a recruitment target; and the pre-registration is lodged. Fieldwork does not open until all four are true.

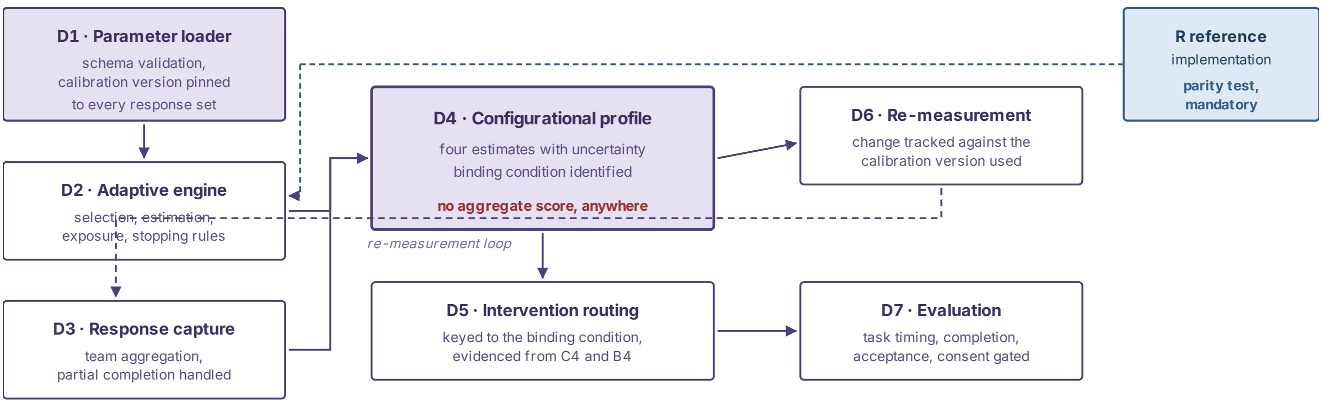
8 Workstream D: The CLEAR-DST Artefact

Built on pilot parameters now, swapped to calibrated parameters later, deployed only after clearance

7 STAGES	**B7** NEEDS ONLY PILOT PARAMETERS TO START	**0** AGGREGATE SCORES, EVER	**2** BLOCKING CLEARANCES BEFORE DEPLOYMENT

Build against pilot parameters now and swap in the calibrated parameters later. The adaptive engine, the interface, the profile logic and the evaluation instrumentation do not need final item parameters to be built correctly, and waiting for fieldwork would put the entire artefact on the critical path for no benefit. What the artefact does need before it meets a real participant is an ethics amendment and a settled institutional intellectual property position, and both of those are started early precisely because neither is under the candidate's control.

> **TWO BLOCKING CONDITIONS**
>
> The artefact may be built freely and deployed to no one. Before it is put in front of a single real participant, the ethics approval must cover the artefact evaluation specifically, and the institutional position on intellectual property in university-developed software must be settled in writing. Both are pursued from Stage B1 and Stage D1 respectively, because both take months and neither can be compressed at the end.



BLOCKING BEFORE ANY REAL PARTICIPANT: ethics amendment covering artefact evaluation · institutional intellectual property position settled in writing

Figure 8 – Artefact architecture. The dashed line from the R reference implementation is the cross-language parity test, which is a mandatory component rather than a quality improvement.

D1 Parameter contract and loader

Depends on B7 · The only coupling between repositories

INSTRUCTIONS

1. Publish the schema for `item-parameters.json` before writing either producer or consumer, and version the schema itself. The file carries: calibration version, calibration date, model type,

sample description, provisional or final status, and per item the dimension, discrimination and category thresholds.

2. Validate on load and fail loudly. A parameter file that does not match the schema stops the application; it does not degrade to a default. Silent fallback in a psychometric engine produces plausible numbers from wrong parameters, which is worse than an error.
3. Record the calibration version against every response set at the moment of administration, not at the moment of reporting. A profile computed under one calibration and re-displayed under another is not comparable to itself.
4. Write the swap procedure as a documented, tested operation: how the file is replaced, what happens to response sets collected under the previous version, and how the interface communicates that earlier and later measurements are not directly comparable.
5. Begin the institutional intellectual property enquiry at this stage. The parameter file is the boundary between research output and product, and the ownership question becomes concrete the moment it crosses.

KEY DELIVERABLES

- [ ] Published, versioned schema for the parameter file
- [ ] Loader with validation that fails closed, with tests for every malformed case
- [ ] Calibration version recorded against every response set at administration time
- [ ] Documented and tested parameter swap procedure
- [ ] Intellectual property enquiry opened, with a written record of the position

D2 Adaptive engine and the parity test

Depends on D1 · Highest technical risk in Workstream D

INSTRUCTIONS

1. Implement maximum information item selection, ability estimation, exposure control and content balancing across the four dimensions. Content balancing is not optional here: an unconstrained algorithm will spend its budget on whichever dimension is most informative and return no estimate for another.
2. Express stopping rules as standard error targets with a maximum length, configurable rather than hard-coded, and read from configuration so that the C6 decision on precision can be applied without a code change.
3. Build the cross-language parity test as a first-class part of the test suite. Identical simulated response vectors are run through the TypeScript engine and the R reference implementation, and the test asserts agreement on ability estimates, standard errors and item selection sequence to a stated tolerance.
4. Run the parity test on every change to either implementation, and fail the build on divergence. A psychometric engine that silently diverges from its reference is worse than no engine at all, because the results look plausible and nothing signals otherwise.
5. Cover the degenerate cases in tests: all responses at one extreme, a single response, an abandoned session, and a dimension in which no item has been administered. Each has a defined behaviour and none of them is a crash or a fabricated estimate.

KEY DELIVERABLES

- [ ] Adaptive engine with configurable selection, estimation, exposure and stopping
- [ ] Content balancing guaranteeing an estimate for every dimension
- [ ] Cross-language parity test in continuous integration, failing the build on divergence
- [ ] Documented tolerance with the justification for its value
- [ ] Degenerate case tests, each with defined behaviour

GATE D2

Parity holds across the full simulated response corpus from Stage C6, including the degenerate cases, and the test runs automatically. Parity established once by hand is not parity; it is a coincidence with a date on it.

D3 Response capture and team aggregation

Depends on D2 · Carries the confidentiality rule

INSTRUCTIONS

1. Model the data so that an individual response set belongs to a person, a team, an organisation and a measurement occasion, and so that the team-level profile is computed rather than stored as an editable value.
2. Implement the same aggregation logic the analysis uses, and test it against the R implementation on the same simulated data. Two different aggregation rules between the research and the product is a silent inconsistency that would surface as an unexplained discrepancy in the evaluation study.
3. Handle partial completion explicitly. Define the minimum number of respondents below which a team profile is not computed, and enforce it in the data layer rather than in the interface, so that it cannot be bypassed.
4. Enforce the confidentiality floor at the same level: below the minimum number of respondents, no team-level output is produced or displayed to anyone, including an administrator. This is an ethics commitment made in the consent form, so it is enforced in code rather than in policy.
5. Store no free text in the first release. Free text is the fastest route to an inadvertent identification and to a data protection problem, and nothing in the diagnostic requires it.

KEY DELIVERABLES

- [ ] Data model supporting longitudinal re-measurement by team and occasion
- [ ] Aggregation logic tested against the R implementation on identical inputs
- [ ] Minimum respondent rule enforced in the data layer
- [ ] Confidentiality floor enforced in code, with tests attempting to bypass it
- [ ] No free-text capture in the release used for evaluation

D4 Configurational profile output

Depends on D3 · The theoretical commitment made visible

This is where the thesis becomes an interface. Everything the argument claims about measurement error is either enforced here or abandoned here.

INSTRUCTIONS

1. Display four dimension estimates, each with its uncertainty shown rather than implied. An estimate presented without its interval invites a precision the data does not support, and with four to five respondents per team the intervals will be wide enough to change how a reader acts.
2. Identify the binding condition explicitly, using the bottleneck logic from Stage C4 rather than by picking the lowest score. The lowest score and the binding condition are not the same thing, and conflating them would reintroduce the ranking logic the thesis argues against.
3. Display no aggregate readiness score, no overall percentage, no letter grade and no composite index, at any point in the interface, in any export, or in any notification. Guardrail G7 is enforced by a test that fails if a composite is computed anywhere in the codebase.
4. Present the profile so that two organisations with the same mean and different binding conditions are visibly different. If the interface cannot show that difference, it has not represented the theory.
5. State the calibration version and the number of respondents on every profile view, since both bound what the profile can be read to mean.
6. Write the interpretation guidance into the interface rather than into a separate document. A diagnostic whose output requires a manual will be misread in the field.

KEY DELIVERABLES

- [ ] Four-dimension profile view with uncertainty displayed
- [ ] Binding condition identified through bottleneck logic, not by minimum score
- [ ] Automated test asserting that no composite score exists anywhere in the codebase
- [ ] Profile view showing calibration version and respondent count
- [ ] Embedded interpretation guidance
- [ ] Two worked example profiles with equal means and different binding conditions

D5 Intervention routing

Depends on D4, C4 · Answers RQ4.2

INSTRUCTIONS

1. Build the intervention library keyed to the four intervention classes from the partition principle, so that each binding condition routes to a distinct class of action rather than to a general improvement list.
2. Attach evidence to every intervention: what it is, what it is expected to change, and the source of that expectation, whether from the review corpus, the practitioner interviews, or the configurational analysis. An intervention without a stated basis is advice, and the artefact is not in the advice business.

3. Distinguish clearly between interventions with empirical support and those derived from practitioner accounts, and label them differently in the interface. Overstating the evidence base of an intervention is the artefact's version of a fabricated citation.
4. Prioritise by the bottleneck ordering from Stage C4 together with implementation feasibility, and expose the reason for the ordering rather than only the result.
5. Where the analysis provides no evidenced intervention for a binding condition, say so in the interface. A gap honestly displayed is more useful than a generic recommendation.

KEY DELIVERABLES

- [ ] Intervention library, keyed to the four classes, with an evidence field on every entry
- [ ] Evidence-strength labelling distinguishing empirical from practitioner-derived
- [ ] Prioritisation logic exposing its own reasoning
- [ ] Explicit handling of conditions for which no evidenced intervention exists

D6 Re-measurement and change tracking

Depends on D4

INSTRUCTIONS

1. Support successive administrations to the same team, with the composition of the responding group recorded at each occasion, because a change in the profile and a change in who answered are easily confused.
2. Compute change per dimension with an interval, and never present a change as meaningful when it falls inside the measurement error. This is where an adaptive instrument with wide intervals will most tempt over-interpretation.
3. Block comparison across calibration versions, or where it is permitted, mark it explicitly and state the limitation in the view itself.
4. Record the interventions undertaken between occasions, so that the re-measurement has something to be interpreted against. Without that record the change data cannot support any claim about what caused it.

KEY DELIVERABLES

- [ ] Longitudinal profile view with respondent composition recorded per occasion
- [ ] Change estimates with intervals, and suppression of changes within measurement error
- [ ] Cross-calibration comparison blocked or explicitly flagged
- [ ] Intervention record linked to the interval between occasions

D7 Evaluation instrumentation

Depends on D4 · Blocks Workstream E

INSTRUCTIONS

1. Instrument the measures the evaluation study needs: task completion, time on task, abandonment point, the acceptance measures, and whatever behavioural trace the evaluation design specifies. Decide these with Stage E1 rather than instrumenting first and designing the study around whatever was captured.

2. Gate all evaluation telemetry behind explicit consent, separate from consent to complete the diagnostic. A participant who agreed to answer questions has not thereby agreed to have their interaction recorded.
3. Make the telemetry inspectable by the participant and deletable on request, and build the deletion path rather than describing it.
4. Capture the artefact version and the calibration version against every evaluation session, so that evaluation results can be attributed to a specific build.
5. Instrument the decision-change measure that RQ4.3 requires: what the participant intended to do before seeing the profile, and what they intended after. Retrofitting this measure is not possible once the evaluation has run.

KEY DELIVERABLES

- [ ] Evaluation telemetry specified jointly with Stage E1
- [ ] Separate consent gate for telemetry, with a tested deletion path
- [ ] Artefact and calibration version recorded per session
- [ ] Before-and-after decision-intent capture for RQ4.3

GATE D7 · WORKSTREAM D DEFINITION OF DONE

A deployable diagnostic running on pilot parameters, with the TypeScript engine verified against the R reference in continuous integration, a documented and tested parameter swap procedure, an automated test proving no aggregate score exists, and evaluation instrumentation ready. Deployment to real participants waits on the two blocking clearances and on nothing else.

9 Workstream E: Evaluation, Publication and Thesis

Turning a working artefact into a defended contribution

5	5	7	E3
STAGES	PAPERS, AND NO MORE	THESIS CHAPTERS	WHERE THE CONTRIBUTION IS STATED

A design science thesis is judged on the knowledge it produces, not on the software it leaves behind. The artefact is the vehicle. The contribution is the set of design principles it demonstrates, and a thesis that describes the build in detail and states the principles in a paragraph has inverted its own argument. Stage E3 exists to prevent that, and it is scheduled before the final write-up rather than during it.

E1 Evaluation strategy and design

Depends on D4 · Specifies D7

Evaluation is planned as a sequence rather than as a single event at the end. Formative evaluation improves the artefact; summative evaluation establishes what it achieves. Conflating them produces an evaluation that is neither.

INSTRUCTIONS

1. Choose and justify the evaluation strategy explicitly against an established framework, stating why this artefact, in this setting, with these risks, calls for the balance of formative and summative work that has been chosen. A strategy asserted without justification is the first thing a design science reviewer will question.
2. Run formative evaluation first, with a small number of practitioners, on the artefact as it stands. The purpose is to find what is misunderstood, not to measure satisfaction, and the output is a change list.
3. Design the summative study around the claim the artefact makes. The claim is that identifying the binding condition changes what an organisation decides to do. The study must therefore capture intended action before and after exposure to the profile, which is why Stage D7 instruments it.
4. Specify the comparison. Against no diagnosis, against a conventional maturity-style score, or within-participant before and after. A comparison against a maturity-style presentation of the same underlying data is the most informative option and is the one that connects the evaluation to H6.
5. Define the criteria in advance: what result would count as the artefact working, and what result would count as it not working. Both must be statable before any participant is recruited.
6. Include practitioners who would use the artefact and practitioners who would be measured by it. These are different populations with different concerns, and an evaluation drawn only from the first will miss the objection that matters most.

7. Specify the qualitative component. Numbers on task completion will not explain why a profile changed a decision, and the explanation is the part that becomes a design principle.

KEY DELIVERABLES

- [ ] Evaluation strategy with a written justification against the framework used
- [ ] Formative evaluation protocol and change list
- [ ] Summative study design with a specified comparison
- [ ] Pre-stated success and failure criteria
- [ ] Participant specification covering users and those measured
- [ ] Telemetry specification handed to Stage D7
- [ ] Ethics amendment lodged, covering the evaluation as designed

E2 Evaluation execution

Depends on E1, D7 · Blocked until both clearances are in place

INSTRUCTIONS

1. Confirm both blocking clearances in writing before recruiting: the ethics amendment covering the evaluation, and the settled intellectual property position. Record the confirmations in the repository.
2. Run the formative round, apply the change list, and record which changes were made and which were declined with reasons. The declined list is as informative as the applied list and belongs in the paper.
3. Freeze the artefact version for the summative round and record it. An artefact changing during its own evaluation cannot be evaluated.
4. Run the summative study to the pre-stated criteria, and report the result against those criteria whether or not it is favourable. An evaluation that can only succeed has not evaluated anything.
5. Collect the qualitative component alongside, and analyse it for the mechanism rather than for endorsement. The question is why the profile did or did not change a decision, not whether participants liked it.
6. Record every technical failure encountered in use. These are findings about the design, and a design science paper that reports no difficulties will not be believed.

KEY DELIVERABLES

- [ ] Written confirmation of both clearances, dated and filed
- [ ] Formative round results with applied and declined changes
- [ ] Frozen artefact version recorded against the summative study
- [ ] Summative results reported against the pre-stated criteria
- [ ] Qualitative analysis of the decision-change mechanism
- [ ] Technical failure log

E3 Design principles and contribution statement

Depends on E2 · Where the thesis is won or lost

The design principles are the transferable knowledge. They must be statable by someone who will never see the software, and applicable to a diagnostic problem that is not this one.

INSTRUCTIONS

1. Write each principle in a consistent form: for a given class of problem, in a given context, a designer should do a particular thing, in order to achieve a particular effect, because of a particular mechanism. A principle without the mechanism clause is a recommendation.
2. Derive each principle from evidence produced by the programme, and cite that evidence. A principle that could have been written before the research began does not belong in the contribution.
3. State the boundary conditions. A principle that claims to hold everywhere is untestable, and naming where it does not hold strengthens rather than weakens the claim.
4. Position the contribution explicitly against the established framing of design science contribution types, and say which type this thesis makes. Leaving that judgement to the examiner is a decision, and it is the wrong one.
5. Write the contribution statement so that it survives separation from the artefact. If the software were deleted tomorrow, what would remain that another researcher could use?

KEY DELIVERABLES

- [ ] Design principles in consistent form, each with a mechanism and boundary conditions
- [ ] Evidence trace from each principle to the finding that produced it
- [ ] Explicit positioning of the contribution type
- [ ] Contribution statement that stands without the software

CANDIDATE PRINCIPLES THE PROGRAMME IS ON COURSE TO SUPPORT

Drafting these early sharpens the analysis that must support them. Three are already visible from the design. That a diagnostic which returns a configurational profile changes a different decision from one that returns a rank, and therefore that composite scoring is a design failure rather than a simplification. That the object of measurement, not the dimension set, is what dates an organisational instrument, and therefore that object-specific items should be separable from the core scale. That a psychometric engine reimplemented outside its reference environment requires a parity test as a structural component, because the failure mode is plausible output rather than an error. Each of these is falsifiable by the evaluation, which is what makes them principles rather than opinions.

E4 Publication sequencing

Runs continuously from Stage A6

Review timelines in this field are long enough that submission order determines whether the thesis waits on the papers. Papers 1 and 2 are submitted while the empirical work is still running, precisely so that they are not on the critical path.

Table 9 – Output plan, with dependencies and the earliest point at which each can be submitted.

Ref.	Content	Depends on	Earliest submission	Thesis chapter
P1	Systematic review and measurement appraisal of culture, innovation readiness and artificial intelligence readiness instruments	Workstream A only	On completion of A6	Chapter 2
P2	Conceptual: culture as the conversion mechanism, and where the constraint binds when the object is machine-derived output	A6, B4	On completion of B4	Chapter 3
P3	Instrument development and validation, with the configurational test of the culture-to-innovation relationship	B7, C2 to C5	After fieldwork and analysis	Chapters 4 and 5
P4	Design and evaluation of the CLEAR-DST artefact, with the design principles	C6, D, E2, E3	After the evaluation study	Chapter 6
P5	The artificial intelligence conversion test: capability, culture and practice change, with the decision model	C4, C5	With or shortly after P3	Chapters 5 and 6

INSTRUCTIONS

1. Submit Paper 1 as soon as Gate A6 closes. It depends on nothing downstream and every month it waits is a month of review time that cannot be recovered later.
2. Select target journals per paper before writing rather than after, since scope, structure and length constraints differ enough to make retrofitting expensive. Record the target and the fallback for each.
3. Check each target's position on the methods used. Not every construction management outlet publishes set-theoretic work, and discovering that after submission costs a full review cycle.
4. Agree authorship order and contribution statements with the supervisor at the outset for all five papers, in writing, once. Renegotiating at submission is avoidable friction at the worst possible moment.
5. Keep every paper's numbers flowing from the pipeline. A revision request arriving eighteen months after submission is answerable in an afternoon if guardrail G2 held, and is a week of archaeology if it did not.
6. Present at a domain conference between Papers 1 and 3 to obtain critique on the framework before it is fixed in a journal submission. Conference feedback is cheaper than reviewer feedback and arrives sooner.

KEY DELIVERABLES

- [ ] Target journal and fallback recorded per paper, with a methods-fit check
- [ ] Written authorship and contribution agreement covering all five outputs
- [ ] Submission tracker with dates, status and revision deadlines
- [ ] Conference presentation of the framework before Paper 3 is submitted

E5 Thesis assembly and defence preparation

Depends on all · Final stage

INSTRUCTIONS

1. Assemble the thesis from the same source files that produced the papers, so that a correction made once propagates everywhere. Divergence between a paper and its chapter is a question the examiner will ask and one that has no good answer.
2. Write Chapter 1 last. Its job is to promise exactly what the thesis delivers, and that is only knowable at the end.
3. Write the limitations section with specificity. Not that the study has limitations, but which ones, and what each does to the interpretation. The aggregation reliability question from Stage C3 and the calibration sample question from Stage C6 are named limitations with stated consequences, and naming them first is stronger than being asked about them.
4. Rehearse the four objections the programme already knows are coming: construct redundancy against absorptive capacity, the culture versus climate measurement claim, the feasibility of adaptive administration at this sample size, and whether the artificial intelligence framing is a genuine contribution or a fashionable veneer. Each has an answer built into the design, and the answers are rehearsed rather than improvised.
5. Prepare the reproducibility demonstration. Being able to rebuild any figure in the thesis from raw inputs during the viva converts the reproducibility claim from an assertion into a demonstration, and few candidates can do it.
6. Draft the postdoctoral proposal from the material the prohibition on a sixth output has been accumulating. That material is a proposal, not a loss.

KEY DELIVERABLES

- [ ] Thesis assembled from shared sources, with no divergence from the papers
- [ ] Limitations section naming each limitation and its consequence for interpretation
- [ ] Rehearsed answers to the four known objections
- [ ] Reproducibility demonstration, tested end to end
- [ ] Postdoctoral proposal drafted from deferred material

GATE E5 · PROGRAMME DEFINITION OF DONE

A thesis whose every number traces to a pipeline, whose every item traces to a development log, whose every screening decision traces to a screening log, and whose artefact is verified against its own reference implementation. That chain of traceability is the programme's distinguishing feature and the strongest thing it can put in front of an examiner.

10 Sequencing, Gates and Timeline

What is built when, what blocks what, and the criterion that closes each stage

10.1 Build order

The ordering below is not a preference. Each entry blocks the ones beneath it, and the two long-lead items, ethics and access, start on day one because neither is under the candidate's control and both can absorb months without warning.

Table 10 – Build order with blocking relationships. Items 1 to 9 can all be completed before survey fieldwork opens, which is the point of the sequence.

Order	Work	Blocks	Blocked by
1	Repository scaffold, gitignore verification, dependency lock, pipeline skeleton, thresholds and seeds files	Everything	Nothing
2	Stage B1: ethics application and access negotiation	B3, all fieldwork, E2	Nothing. Start immediately.
3	Stages A1 to A3: protocol, registration, search, deduplication	A4 onward	Item 1
4	Stage C1: the data generator	All of Workstream C	Item 1
5	Stages A4 to A6: screening, appraisal, Paper 1	B5 item inheritance	Item 3
6	Stages B2 to B4: interviews and analysis	B5, Paper 2	Items 2 and 4
7	Stages C2 to C5 on simulated data	C7, and the field analysis	Item 4
8	Stage B5 to B7: item generation, content validity, cognitive testing, pilot	Survey fieldwork, Workstream D	Items 5 and 6
9	Stage C7: feasibility grid and pre-registration	Survey fieldwork opening	Item 7
10	Stages D1 to D4 on pilot parameters	D5 to D7	Item 8
11	Survey fieldwork	The field analysis, C6	Items 8 and 9
12	Stage C6: calibration and adaptive simulation	The parameter swap	Item 11
13	Stages D5 to D7 and the parameter swap	E2	Items 10 and 12
14	Stages E1 to E3: evaluation and design principles	Paper 4, thesis	Item 13

10.2 Indicative timeline

The schedule below is anchored on one fixed point carried from the previous revision, which is that fieldwork opens in the second quarter of 2027. Everything else is derived from the dependency structure. The dates are indicative and must be reconciled against the actual registration date, the funding end date and the submission deadline before this roadmap is treated as a plan of record.

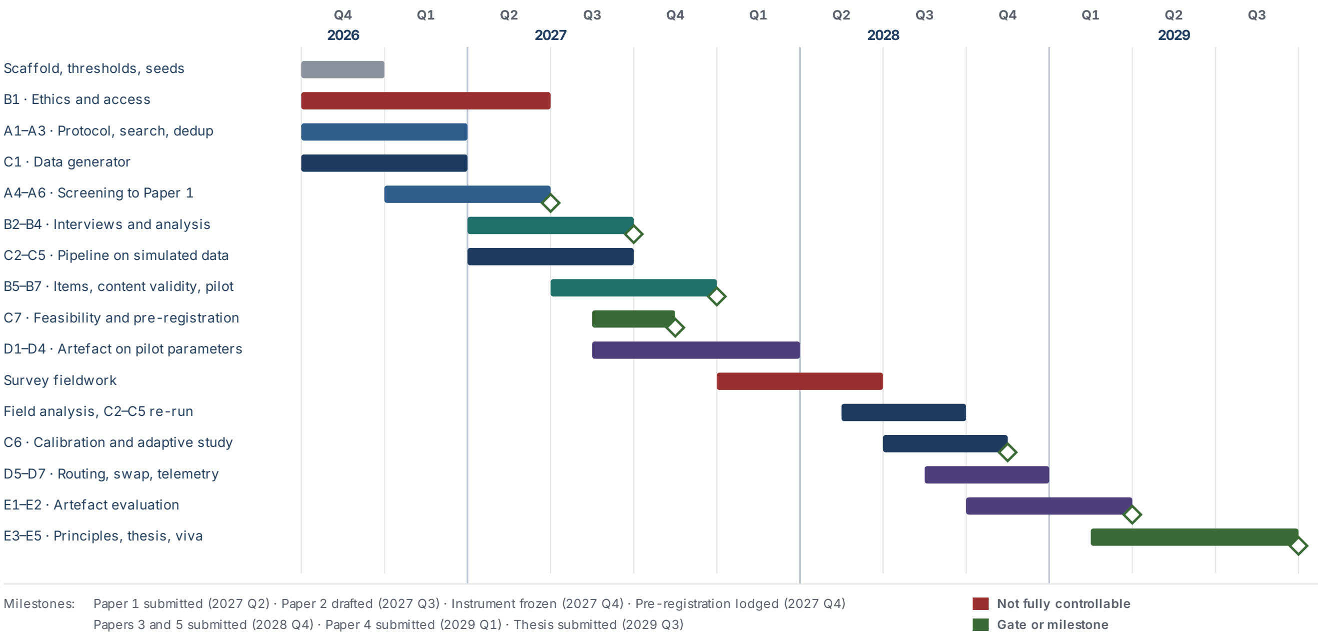


Figure 9 – Indicative programme timeline. The two bars in red are the items least under the candidate's control, and both sit on the critical path. Dates require reconciliation against the registration and funding record before adoption.

10.3 Stage gates

A gate is a criterion, not a review meeting. Work does not pass a gate because the date arrived.

Table 11 – Stage gates and the evidence that closes each.

Gate	Criterion	Evidence that closes it
A1	Two screeners independently reach acceptable agreement on twenty pilot records without conferring	Agreement statistic on the pilot set, with the criteria revision it prompted
A4	The flow diagram regenerates from the screening log and matches the manuscript	Single-command rebuild producing identical counts
A6	The review reproduces from raw exports with no manual step	Clean pipeline run and completed reporting checklist
B1	Ethics approved and access agreements signed for enough organisations to reach the C7 recruitment target	Approval letter and signed agreements, filed
B7	Instrument frozen with full item traceability, and provisional parameters consumed by the artefact	Item development log, and a successful parameter load in the artefact repository
C2	The code recovers the four-factor structure from the four-factor world and the three-factor structure from the three-factor world	Both recovery runs, reported

Gate	Criterion	Evidence that closes it
C7	Full pipeline runs end to end on simulated data; recruitment target derived; pre-registration lodged	Pipeline run log, feasibility grid, timestamped registration
D2	Cross-language parity holds across the full simulated corpus, automatically, on every change	Continuous integration record showing the parity test running and passing
D7	Artefact deployable, no composite score present, evaluation instrumentation ready	Passing composite-absence test, and the telemetry specification signed off against E1
E2	Both clearances confirmed in writing before any real participant is recruited	Ethics amendment and intellectual property position, dated and filed
E5	Every number in the thesis traces to a pipeline, every item to a log, every screening decision to a record	Live reproducibility demonstration from raw inputs

11 Risk Register

What can go wrong, how likely it is, and what has already been built to absorb it

A risk register whose entries have no owner and no mitigation is a document written to be filed. Every entry below names the stage that carries it and the specific thing that has been built into the design to absorb it. Four of these risks are structural, meaning they cannot be eliminated and must be managed and reported.

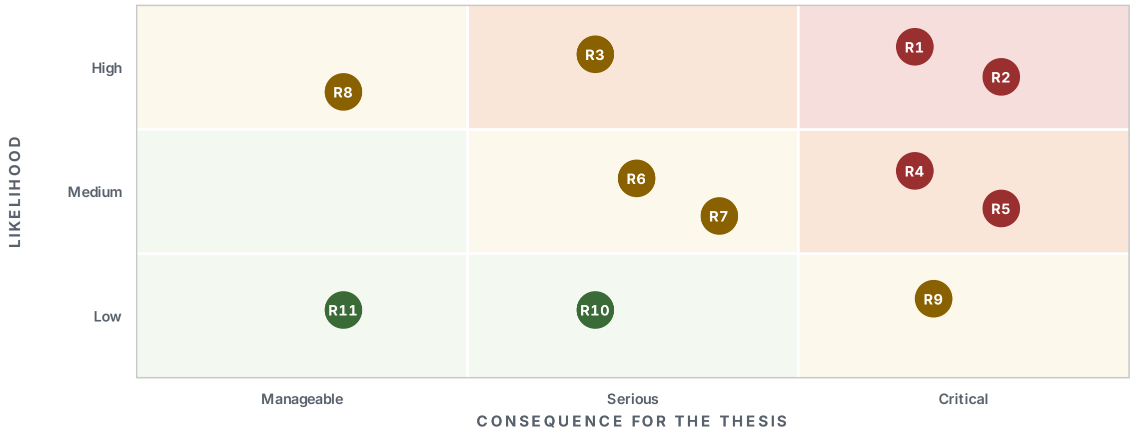


Figure 10 – Risk position at Revision B. The upper right quadrant contains the risks that would change the shape of the thesis rather than delay it, and all four are addressed by design decisions already taken rather than by contingency.

Table 12 – Risk register. Structural risks cannot be eliminated; they are managed, quantified and reported.

Ref.	Risk	Stage	Mitigation already built into the design
R1	**Team-mean reliability is too low to defend aggregation**, because four to five respondents per team bounds it structurally structural	C3	Quantified in advance across the intraclass correlation sweep at Stage C1, mapped at C3 and priced at C7. Three responses pre-selected: raise respondents per team where access permits, run the structural model at individual level with cluster-robust errors, or report honestly against the interpretive standards and argue on the other two statistics. The decision is recorded in the pre-registration rather than taken after seeing the data.
R2	**The calibration sample is too small for stable graded-response parameters**, and the promised adaptive short form cannot reach the precision claimed structural	C6	The arithmetic is confronted in Stage C6 rather than discovered. Precision is restated as a standard error target instead of an item count, a multidimensional model is fitted to borrow information across correlated dimensions, and the fallback is a sixteen to twenty item adaptive form which remains a reduction of more than half.
R3	**Recruitment falls short of the team target**, leaving the multilevel model underpowered	B1, C7	Access is item two in the build order and starts on day one. The design-effect-adjusted single-level fallback is built and tested on simulated data at Stage C5, so switching to it is a sample-size decision made against a pre-registered threshold rather than an improvisation.

Ref.	Risk	Stage	Mitigation already built into the design
R4	**The four-factor structure does not hold** and the social conditions collapse into one	C2	The three-factor rival is generated, specified and pre-registered as a competing model rather than treated as a fallback. The consequence for the artefact, three intervention classes rather than four, is worked out in advance so that the finding is reportable rather than disruptive.
R5	**The instrument is challenged as a climate measure presented as a culture measure**	C2, thesis	The measurement claim is stated narrowly and defended on its own terms in Chapter 3, the composition model is named, and the qualitative phase carries the deeper cultural account that survey items cannot reach. Conceding the distinction is stronger than defending the broader claim.
R6	**Construct redundancy against realised absorptive capacity**	B5, C2	A short validated absorptive capacity scale is carried in the instrument from Stage B5 and discriminant validity is tested against it directly at Stage C2. The objection is answered with data rather than with argument, at a cost of eight items.
R7	**Common method variance inflates the culture-to-innovation relationship**	B5, C2, C5	Separate informant for the team-level outcome, temporal separation between predictor and outcome blocks, and a marker variable carried in the instrument. Procedural remedies designed in rather than statistical remedies applied after.
R8	**Set-theoretic analysis is criticised for arbitrary calibration**	C4	Anchors are declared in a version-controlled configuration file with a written justification for each, and the robustness protocol varies anchors and thresholds across a stated range with the sensitivity reported.
R9	**Ethics or intellectual property clearance is delayed past the evaluation window**	B1, D1, E2	One ethics application covers all three phases rather than three sequential applications. The intellectual property enquiry opens at Stage D1, long before the artefact meets a participant. Both are tracked as blocking items rather than administrative tasks.
R10	**Review timelines delay the thesis**	E4	Papers 1 and 2 depend on nothing downstream and are submitted while empirical work runs. The thesis is assembled from shared sources so that it does not wait on acceptance.
R11	**Participant data reaches version control**	B3, all	The ignore rule is verified before any data exists rather than after. Transcripts and raw responses live outside the repository entirely, and the repository holds only coding frames, memos and aggregate output.

THE TWO RISKS THAT MUST BE REPORTED RATHER THAN SOLVED

R1 and R2 are properties of the design at the sample size available, not defects to be engineered away. Both are named in the limitations section with their consequence for interpretation stated, and both are quantified by the simulation work rather than estimated. A thesis that names its structural constraints and prices them is in a stronger position than one that discovers them under examination.

12 Ethics, Data Protection and Intellectual Property

The obligations that constrain what may be collected, stored, reported and deployed

12.1 The employer and employee problem

Access is granted by management and data is given by employees. That asymmetry is the defining ethical feature of this study and a generic consent form does not answer it. An employee who understands that their organisation has sponsored the research, that their manager knows the survey is running, and that their team is small, has good reason to answer carefully rather than honestly. Every design decision below exists to reduce that reason.

1. State in the participant information sheet, in plain terms, that participation is voluntary, that no list of participants or non-participants is provided to the employer, and that no individual response is ever returned to the organisation in any form.
2. Set and publish a minimum number of respondents below which no team-level result is produced, and enforce it in the data layer of the artefact rather than as a reporting convention. Tell participants the number.
3. Suppress any breakdown that would identify an individual by combination of characteristics. A team of four with one female member and one role-specific breakdown has been identified whatever the field labels say.
4. Arrange administration so that a manager cannot observe who is completing the instrument and who is not, and say in the information sheet how that has been arranged.
5. Give participants a route to withdraw that does not require contacting anyone in their organisation, and state the point after which withdrawal is no longer possible because the data has been aggregated beyond recovery.

12.2 Data protection

1. Establish and record the lawful basis for processing before collection, and reflect it accurately in the information sheet. Consent and the public interest basis carry different obligations and different withdrawal consequences, and the choice must be made deliberately with the institution's data protection office rather than assumed.
2. Distinguish pseudonymised from anonymised data honestly throughout. Data holding a re-identification key is pseudonymised and remains personal data under the applicable regime; describing it as anonymous in a consent form is a misrepresentation to participants as well as a compliance failure.
3. Apply data minimisation to the instrument itself. Every demographic field must justify its presence against a specific analytical use, and fields collected because they might be interesting are precisely the fields that enable re-identification.
4. Set the retention period and the destruction point for recordings, transcripts and identifier keys, record them in the data management plan, and diarise them. A retention schedule that is never executed is not a retention schedule.

5. Assess whether the processing triggers a formal impact assessment, taking into account employee data, systematic evaluation of personal aspects, and the artefact's telemetry. Where it does, complete it before collection rather than in parallel with it.
6. Treat the artefact as a separate processing activity with its own basis, its own retention and its own subject-rights routes, since it stores responses beyond the research study and continues after it.

> **INSTITUTIONAL DEPENDENCY**
>
> The lawful basis, the impact assessment trigger and the precise ethics route are institutional determinations, not decisions the candidate makes alone. Confirm each with the institution's research ethics committee and data protection office in writing, and file the confirmations. This roadmap states the obligations to be discharged; it does not substitute for that confirmation.

12.3 Intellectual property

Two questions are settled before the artefact meets a participant, and both are settled in writing. The first is ownership of software developed by a registered candidate using institutional resources, which varies by institution and by funding source and is rarely what people assume. The second is what may be done with the artefact afterwards: whether it can be licensed, commercialised, released openly, or used with organisations outside the study.

1. Open the enquiry at Stage D1, when the parameter file first crosses the repository boundary and the question becomes concrete.
2. Establish the position in writing, covering ownership, permitted use during the study, and permitted use afterwards.
3. Establish separately what participating organisations may do with their own profile data, and state it in the access agreement rather than leaving it to be argued later.
4. Record the licence status of every inherited instrument item from Stage A5. Reproducing copyrighted items without permission is an infringement whether or not the resulting instrument is commercialised.
5. Decide the position on releasing the item parameters and the instrument openly. Open release strengthens the contribution and constrains commercialisation, and the two cannot both be maximised.

13 Prohibitions

The list that does not get suspended when the schedule slips

Each of the following prevents a specific failure that is either irreversible or expensive to reverse. They are listed together because the moment any one of them looks negotiable is the moment the schedule is under pressure, which is exactly when the reasoning behind them is least accessible.

Table 13 – Prohibitions, with the failure each prevents.

Do not	Because
Write results, statistics or citations into any manuscript file directly	A hand-typed number cannot be updated when the analysis changes, and it will not be updated. This is how a manuscript comes to disagree with its own pipeline.
Merge the two repositories	Institutional intellectual property in a research repository is a different question from intellectual property in a product repository. If a task appears to need both, export a parameter file.
Commit anything to `data/raw/`, or any transcript anywhere	Git history is not erasable in practice once shared, and the consent given to participants did not extend to this.
Add an aggregate readiness score to the artefact	It permits exactly the substitution the thesis argues against, and it will be requested by users who find the profile harder to read than a number.
Deploy the artefact to real participants before both clearances	Data collected without clearance is unusable, and the harm to the institutional relationship outlasts the study.
Build the analysis pipeline for the four-condition case only	The instrument may collapse to three dimensions, and the code must be proven to detect that before the data arrives rather than being written under deadline afterwards.
Remove the `SIMULATED` label from any output at any stage	There must exist no path by which a simulated result is mistaken for an empirical one, including by the candidate.
Fit by modification index and report the result as confirmatory	It is detectable, it is common, and it converts a confirmatory analysis into an exploratory one without saying so.
Set fuzzy-set calibration anchors at sample quantiles without a substantive reason	It is the most frequently and most correctly criticised element of published set-theoretic work.
Report a single agreement statistic as sufficient evidence for aggregation	ICC(1), ICC(2) and within-group agreement answer different questions, and reporting only the favourable one is a choice a reader will notice.
Add a measure to the analysis plan after the instrument is frozen	It cannot be recovered at any price once fieldwork has run. Check the plan against the instrument before Stage B7 closes.
Start a sixth research output	The programme produces five. Anything new goes into the postdoctoral proposal, which is where the deferred material has been accumulating for exactly this purpose.

Indicator Blueprint

The content each CLEAR dimension must span, and the item budget that spans it

The blueprint fixes what each dimension must cover before any item is written. It exists so that item reduction at Stage B7 is governed by content coverage rather than by loading magnitude, which is what prevents a dimension ending up with ten near-identical items, high reliability and narrow content.

Table 14 – Indicator blueprint. Item counts are targets for the reduced instrument; the pool written at Stage B5 is roughly three times each figure.

	Facet	Items	What the facet must capture
L · LEADERSHIP · VERTICAL SOCIAL · TEN ITEMS			
L1	Invitation to challenge	3	Whether those with authority actively solicit disagreement, and whether soliciting it is distinguishable from tolerating it.
L2	Response to ideas	4	What happens after an idea is raised: whether it is acknowledged, whether an outcome is communicated, and whether the person hears anything at all.
L3	Risk absorption and sponsorship	3	Whether a manager carries the consequence of a trial that fails, and whether ideas are carried upward beyond the team.
E · ENVIRONMENT · HORIZONTAL SOCIAL · TEN ITEMS			
E1	Interpersonal safety	4	Whether it is safe among peers to be wrong, to ask, and to admit not knowing. Referent is the team throughout.
E2	Voice norms	3	Whether raising a concern is ordinary or exceptional, and whether it is done openly or through back channels.
E3	Error and blame response	3	What the team does when something goes wrong, and whether the response locates a cause or a person.
A · ARCHITECTURE · STRUCTURAL · TEN ITEMS			
A1	Decision rights	3	Whether it is knowable who decides, and whether that person is reachable from where the idea originated.
A2	Authorisation route	4	Whether a route exists from proposal to authorised change, how long it takes, and whether it is used.
A3	Governance and contract interface	3	Whether contractual and governance arrangements permit a change to be made, and who carries the risk of making it.
R · RESOURCES · MATERIAL · TEN ITEMS			
R1	Time and budget slack	4	Whether there is any capacity beyond delivery, and whether attempting an improvement is chargeable to anything.
R2	Skill and development	3	Whether people have, or can acquire, the capability the change would require.

	Facet	Items	What the facet must capture
R3	Data and tooling access	3	Whether the data and tools needed are available to the people who would use them, including artificial intelligence tooling.
OBJECT-SPECIFIC BLOCK · SCORED SEPARATELY, NEVER FOLDED INTO THE DIMENSION SCORES			
X1–X4	Artificial intelligence object substitution	4	One item per dimension, asking the same condition about machine-derived output rather than about a colleague's proposal. Kept separable so that the core instrument does not date as the technology moves.
CARRIED MEASURES · REQUIRED BY THE ANALYSIS PLAN, FIXED AT STAGE B5			
—	Psychological safety	—	Established published scale, referent-shifted, carried whole so that the mediation in H2 is estimated on a validated measure rather than on a subset of Environment.
—	Innovative work behaviour	—	Established published scale covering idea generation, promotion and realisation. Team-level outcome obtained from a separate informant.
—	Realised absorptive capacity	—	Short validated scale, carried solely to establish discriminant validity against the construct redundancy objection.
—	Marker variable	—	Theoretically unrelated to every construct in the model, carried for the method variance assessment at Stage C2.
—	Artificial intelligence capability index	—	Inventory of tooling, data access and skill actually held. An inventory rather than a perception, so that H5 has something to interact with that is not itself a rating.
—	Technology maturity score	—	Constructed from the review corpus as the conventional rival predictor, so that H6 is a real contest.

LENGTH DISCIPLINE

The carried measures make the instrument considerably longer than forty items, and Stage B6 measures the consequence. Where the total exceeds what a site-based respondent will complete without satisficing, the reduction comes from the CLEAR pool and never from the carried measures, because a shortened published scale is no longer the published scale and loses the comparability it was carried for.

Measurement Appraisal Rubric

The six criteria applied to every instrument in the review corpus

Each instrument is scored on every criterion, and every score carries an evidence field recording what justified it. The rubric is fixed at Stage A1, before any record is read, and is reproduced verbatim in the registered protocol.

Table 15 – Six-criterion appraisal rubric with scoring anchors.

	Criterion	Score 0	Score 1	Score 2
C1	**Construct definition** Is it clear what is being measured?	No definition, or a label used in place of a definition.	A definition is given but does not distinguish the construct from adjacent ones.	A definition is given that would let a reader decide whether a candidate item belongs.
C2	**Theoretical anchoring** Where does the structure come from?	Dimensions asserted, or derived only from exploratory analysis of the development sample.	Dimensions drawn from prior literature without a stated theory of why these and not others.	Dimensions derived from a stated theory that explains the partition and predicts relationships.
C3	**Object of measurement** What do the items actually ask about?	Predominantly the presence of technology or the existence of policy.	A mixture of presence and perception, not distinguished by the authors.	Perceived conditions or observed behaviours, with the object stated and consistent.
C4	**Level of analysis** Whose property is being measured?	An organisational property claimed from individual responses with no aggregation argument.	Aggregation performed with partial justification, or a composition model implied but not named.	Composition model named, aggregation statistics reported, referent consistent with the model.
C5	**Psychometric evidence** Has structure been tested?	None, or internal consistency alone.	Factor structure examined, but exploratory only or in the development sample only.	Confirmatory evidence in an independent sample, with discriminant validity and, where relevant, invariance.
C6	**Predictive evidence** Does it relate to anything?	No relationship to any outcome reported.	Cross-sectional association with a self-reported outcome from the same respondents.	Association with an outcome measured independently, or over time.

HOW THE RUBRIC IS USED

The scored matrix is the review's primary result and its central claim is expected to sit in criterion C3. If the corpus scores well on C5 and badly on C3, the field has been measuring the wrong object carefully, which is a more interesting and more defensible finding than a general complaint about rigour. The rubric must permit that pattern to be visible, and it must equally permit the opposite.

Simulation Parameter Grid

What the generator varies, and what the feasibility analysis sweeps

The generator's parameter interface is fixed here so that Stage C7 has a defined grid rather than an ad hoc one, and so that the feasibility results are reproducible from the parameter file alone.

Table 16 – Generator parameters. Values marked as swept are varied across the Stage C7 design analysis; values marked as fixed are held at the planned design.

Parameter	Status	Purpose in the design
STRUCTURE		
Number of teams	swept	Determines whether the multilevel model is estimable and whether the configurational analysis has enough cases. Swept across the plausible recruitment range to locate the abandonment threshold.
Respondents per team	swept	Drives team-mean reliability directly. This is the parameter that decides risk R1, and the sweep quantifies what an additional respondent per team is worth.
Dimensional world	swept	Four-condition and three-condition generating structures. Both must be recovered correctly by the Stage C2 code.
Items per dimension	fixed	Held at the blueprint target so that the simulation matches the instrument that will be fielded.
Response categories	fixed	Five, matching the instrument. Category collapse is examined at Stage C6 rather than simulated here.
VARIANCE AND EFFECT		
Intraclass correlation	swept	Swept across the range plausible for organisational perception data, to map where aggregation is defensible and where it is not.
Culture to innovation effect	swept	Sets the detectability of H1 and drives the power surface.
Mediation path strengths	swept	H2 requires two indirect paths to be distinguishable from each other, which is a harder test than either being non-zero.
Capability interaction size	swept	H5 is a cross-level interaction and will require substantially more information than the main effects. The sweep establishes how much more.
Inter-dimension correlation	swept	Governs both discriminant validity at Stage C2 and the efficiency gain available from a multidimensional adaptive form at Stage C6.
NUISANCE AND REALISM		
Response compression	on by default	Observed in the pilot. Shrinks the usable response range and depresses variance, which affects reliability, calibration and agreement statistics alike.

Parameter	Status	Purpose in the design
Social desirability shift	on by default	Respondent-level additive shift correlated with the marker variable, so that the method variance assessment at Stage C2 is exercised against something real.
Item-level missingness	on by default	Exercises the missing data handling before it meets real data.
Partial team completion	on by default	Teams returning two, three, four and five members. The aggregation and multilevel code must handle all of them, and the minimum-respondent rule must be exercised.
CONTROL		
Seed	required	Drawn from `config/seeds.yml`. No stochastic operation runs without one.
Replications per cell	fixed	Set high enough that the Monte Carlo error on a detection rate is small relative to the differences being read from the grid, and stated in the pre-registration.
Metadata label	required	Every generated object carries its parameters, its seed and the word `SIMULATED`, propagated automatically into every downstream artefact.

Appendix D

Reporting Checklists

What each analysis must report, decided before the analysis is run

Each list below is what the corresponding stage reports regardless of the result. Deciding this in advance is what prevents selective reporting, and it is cheap to do now.

MEASUREMENT MODEL · STAGE C2

- [ ] Estimator, and the reason for treating items as ordinal
- [ ] Both competing structures, with the comparison statistic
- [ ] Robust fit indices, with cutoffs cited to a source
- [ ] All standardised loadings, including the weak ones
- [ ] Every departure from the specified model, with justification
- [ ] Categorical omega with intervals, per dimension
- [ ] Full discriminant matrix, with inferential bounds
- [ ] Discriminant result against absorptive capacity
- [ ] Marker-variable method variance assessment

INVARIANCE AND AGGREGATION · STAGE C3

- [ ] Full invariance sequence with the ordinal steps named
- [ ] Change criteria used and their source
- [ ] Any partial invariance and what it permits
- [ ] ICC(1), ICC(2) and agreement, all three
- [ ] Agreement under more than one null distribution
- [ ] Between-team variance test
- [ ] The aggregation decision and its stated fallback

CONFIGURATIONAL · STAGE C4

- [ ] Every calibration anchor with its justification
- [ ] Cases at the crossover point, counted
- [ ] Necessity before sufficiency, with effect sizes and tests
- [ ] Bottleneck table
- [ ] Truth table with frequency and consistency thresholds
- [ ] Extent of limited diversity, stated honestly
- [ ] All three solution types, core and peripheral marked
- [ ] Robustness across anchors and thresholds
- [ ] Solution for the negated outcome
- [ ] Parallel solution on the artificial intelligence outcome

STRUCTURAL · STAGE C5

- [ ] Variance partition before any predictor
- [ ] Level of every effect, stated unambiguously
- [ ] Indirect effects with Monte Carlo intervals
- [ ] Competing indirect paths in one model
- [ ] Interaction and simple slopes for H5
- [ ] Capability main effect with the interaction present
- [ ] Nested comparison for H6 on the pre-specified criterion
- [ ] Models with and without controls
- [ ] Whether the fallback was used, and why

ITEM RESPONSE · STAGE C6

- [ ] Assumption checks per dimension
- [ ] Item fit statistics and threshold ordering
- [ ] Information functions and the precision range
- [ ] Calibration sample against the requirement, gap named
- [ ] Multidimensional versus per-dimension comparison
- [ ] Adaptive study: precision, exposure, length, trajectory
- [ ] Precision target as a standard error, not an item count

REVIEW · WORKSTREAM A

- [ ] Complete search strings for every source
- [ ] Dates, filters and yields per source
- [ ] Generated flow diagram, all counts derived
- [ ] Agreement on the dual-screened subsample
- [ ] Full-text exclusions with reasons
- [ ] Appraisal scores with evidence on every cell
- [ ] Object-of-measurement tabulation
- [ ] Completed reporting checklist for the target journal

Appendix E

Standing Context File

The condensed form of this roadmap to place at the root of each repository

This roadmap is the reasoning. The file below is the operating instruction derived from it, and it is what an assistant or a collaborator reads before touching the repository. It is kept short deliberately, because a standing context file that is not read in full is not a standing context file.

```
# CLAUDE.md · phd-culture-conversion

## What this is
Doctoral programme on organisational culture as the conversion mechanism for
innovation and artificial intelligence readiness in construction project
environments. Instrument: CLEAR (Leadership, Environment, Architecture,
Resources). Artefact: CLEAR-DST, in a separate repository.
Full reasoning: docs/Corcoran_PhD_Roadmap_RevB.pdf. Read it before acting.

## Critical fact
No primary data exists yet. Survey fieldwork opens 2028 Q1. Everything before
then is built against seeded simulated data and swapped over later. That is
deliberate.

## Absolute rules
G1  Never fabricate. No invented data, results, statistics, citations, DOIs or
    page numbers, in any file, for any reason, including as a placeholder.
    Write TODO_FROM_PIPELINE and stop.
G2  Numbers reach manuscripts only through the pipeline. No hand-typed results.
G3  Simulated data is labelled SIMULATED in the filename and in the metadata,
    always, and the label is never removed.
G4  No participant data in version control. data/raw/ is gitignored at
    creation. No transcripts, names or free text, in either repository.
G5  Everything is seeded from config/seeds.yml and rebuilds with one command.
G6  Every analytical threshold lives in config/thresholds.yml with its source.
G7  The artefact never displays an aggregate readiness score.
G8  British English throughout: analyse, behaviour, organisation, modelling.

## Never
- Merge the two repositories. Export a parameter file instead.
- Build the analysis for the four-factor case only.
- Fit by modification index and report it as confirmatory.
- Add a measure after the instrument is frozen at B7.
- Deploy the artefact to a real participant before ethics and IP clearance.
- Start a sixth research output.

## Workstreams
A  Systematic review of measurement instruments        -> Paper 1, Ch. 2
B  Construct development: interviews, items, pilot     -> Papers 2, 3
C  Analysis pipeline, simulation first                 -> Papers 3, 5
D  CLEAR-DST artefact                                  -> Paper 4, Ch. 6
E  Evaluation, publication, thesis                     -> Ch. 6, 7

## Build order
scaffold -> ethics and access -> review -> generator -> pipeline on simulated
data -> items and pilot -> feasibility and pre-registration -> artefact on
pilot parameters -> fieldwork -> calibration -> parameter swap -> evaluation

## Stack
R: targets, renv, lavaan, semTools, psych, MBESS, mirt, NCA, QCA, quarto.
Artefact: TypeScript, Next.js App Router, PostgreSQL, Prisma.
Verify package interfaces against current releases at setup. Several have
changed signatures across major versions and the failure mode is silent.
```

KEEPING THE TWO IN STEP

When this roadmap changes, the context file is regenerated from it rather than edited separately. Two documents that state the same rules in different words will eventually state them differently, and the one that gets read is not the one that gets updated.